



FAPS

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Soft magnetic materials and manufacturing of lamination stacks

Vorlesung Produktion elektrischer Motoren und Maschinen
Lecture Production of electric motors and machines

The flux in the electric drive is guided by soft magnetic materials.

Soft magnetic materials are used to guide and amplify the magnetic flux in an electrical machine and contribute significantly to energy efficiency due to their function-determining use.

After attending the lecture unit "Soft magnetic materials and manufacturing of lamination stacks", you should know:

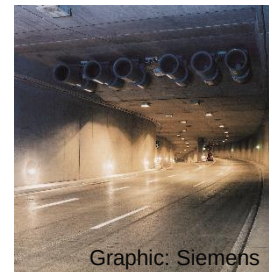
- the essential properties of soft magnetic materials and their processing
- the manufacturing process and classification of electrical steel and the insulation coating
- the most important processing methods for cutting and stacking of electrical steel and
- the production of soft magnetic composites.



Graphic: Omron



Graphic: ZF Friedrichshafen

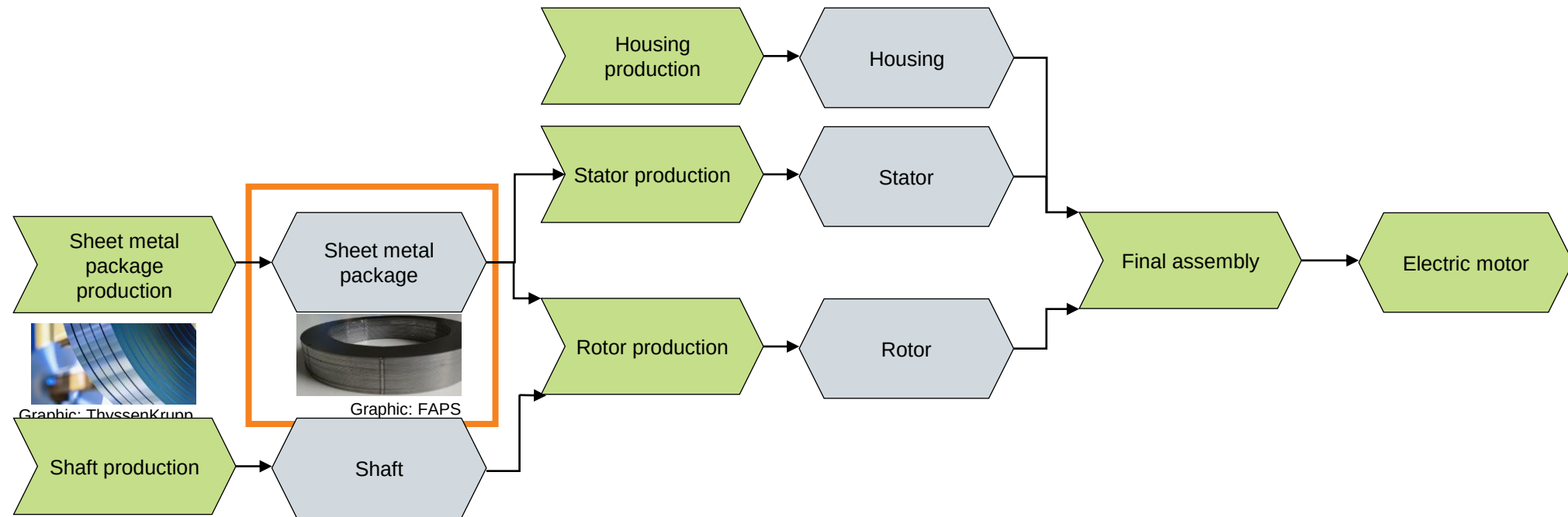


Graphic: Siemens



Graphic: Kuka

The process chain for manufacturing an electric motor comprises a large number of process steps and material elements.



Magnetic materials can be divided into soft magnetic and hard magnetic.

Magnetic materials

Soft magnetic

Use among others:

- Electric drives (stator / rotor)
- Generators
- Contactors, relays, transformers

Parameters:

- High saturation flux density
- Narrow hysteresis
- Low coercivity

Materials:

FeNi, FeCo, FeSi

Lecture unit
2

Soft magnetic rotor and stator of a reluctance machine

Graphic: ZF Friedrichshafen AG



Hard magnetic

Use among others:

- PM rotors
- Switching magnets
- Holding magnets
- Medical diagnosis
- Sensors

Parameters:

- Wide hysteresis
- High coercivity
- High retentivity

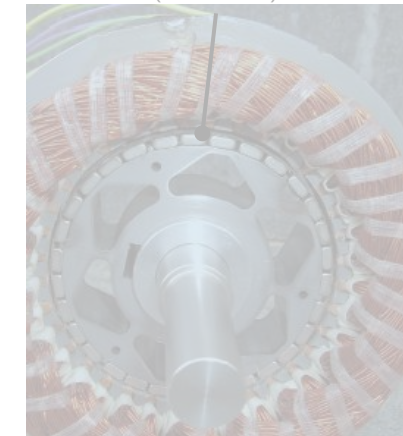
Materials:

Hard ferrites, NdFeB, SmCo, plastic-bonded, AlNiCo

Lecture unit
3

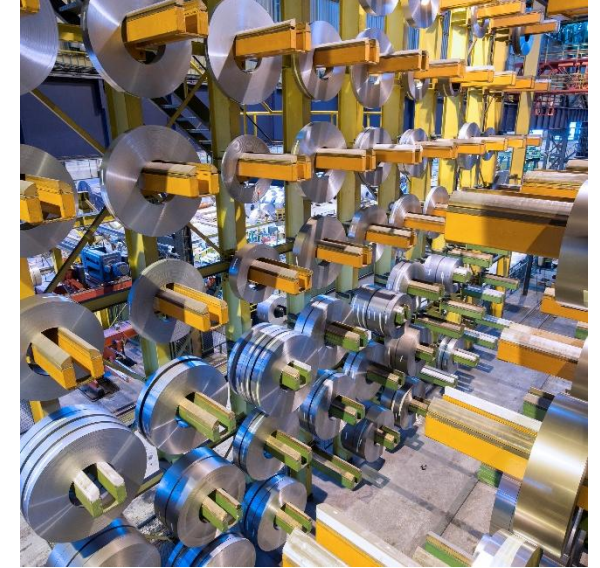
Hard magnetic rotor of a permanent magnet synchronous machine

Graphic: Swiss Federal Office of Energy SFOE (Switzerland)



Lecture unit 2 provides an overview of the classification and manufacturing of soft magnetic materials.

- **Overview and classification of soft magnetic materials (especially electrical steel)**
- Production of electrical steel
- Insulating coatings for electrical sheets
- Processing of electrical steel (cutting, packaging)
- Alternative soft magnetic core materials

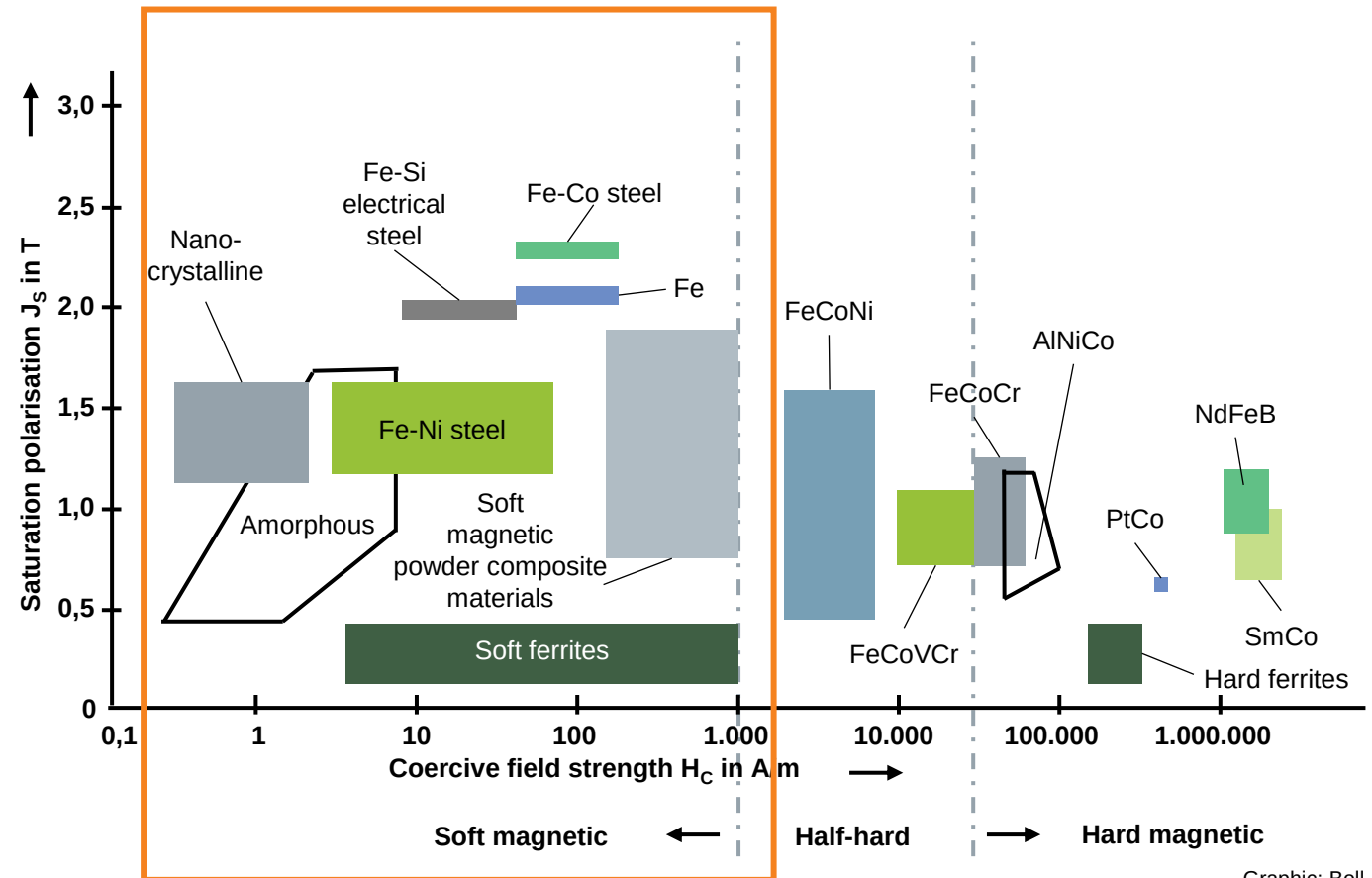


Graphic: ThyssenKrupp

Soft magnetic materials can be categorized according to saturation polarization and coercivity field strength.

Classification of soft magnetic materials:

- Classification according to saturation polarization J_s and coercive field strength H_C
 - The saturation polarization J_s is the highest achievable magnetic polarization J in a magnetic material at a given temperature
 - The coercive field strength H_C is the magnetic field strength H of the point on the static saturation hysteresis loop for which the magnetic flux density B is zero
- Nanocrystalline and amorphous materials are characterized by low demagnetization losses
- Powder composites have a relatively high electrical resistivity
- Soft magnetic ferrites are used for shell cores and antenna rods
- **Electrical steel is an Fe-Si alloy and is particularly suitable for use in electric drives**



Graphic: Boll

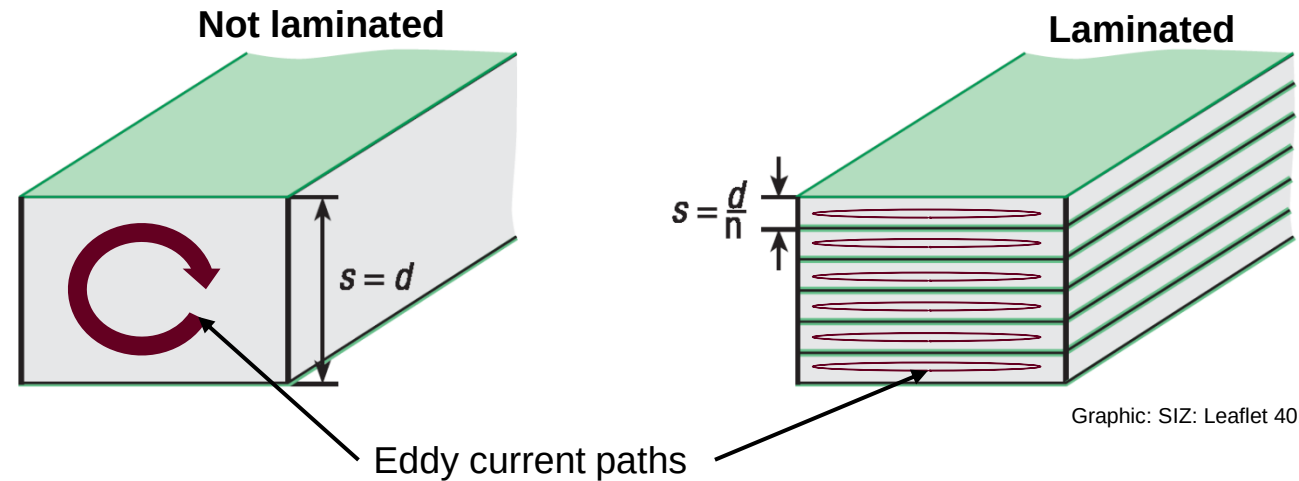
Lamination of magnetic cores is an efficient method to reduce magnetic losses.

Iron losses in electrical machines:

- Static hysteresis and dynamic eddy current losses occur during operation
- Eddy current losses are caused by induced currents and are divided into classic P_{Wk} and anomalous losses P_{Wa}
- Eddy current losses increase quadratically with the sheet thickness and frequency
- Hysteresis losses P_{Hy} occur when the Weiss districts are shifted

Reduction measures:

- Lamination of the magnetic core
- Use of easily magnetizable materials
- Reduction of production-related material damage
- Alloying elements with high specific electrical resistance



Graphic: SIZ: Leaflet 401, SIZ

One loss model is according to Bertotti with homogeneous field curve and sinusoidal flux density curve:

$$P_{Fe} = P_{Hy} + P_{Wk} + P_{Wa} = k_h \cdot f \cdot \hat{B}^2 + k_{Wk} \cdot f^2 \cdot \hat{B}^2 + k_{Wa} \cdot f^{1,5} \cdot \hat{B}^{1,5}$$

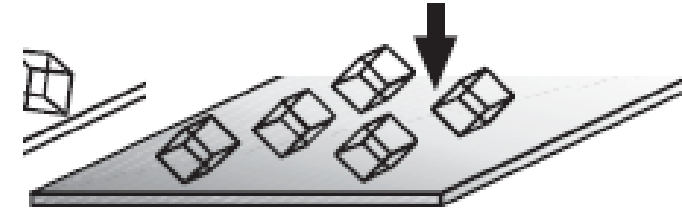
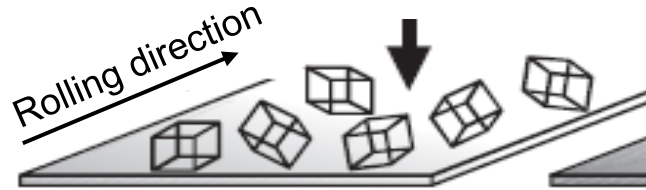
Analytical calculation of eddy current losses:

$$P_W = \frac{\sigma_{Fe} \cdot \pi^2 \cdot d^2 \cdot f^2 \cdot \hat{B}^2}{6 \cdot \rho_{Fe}} \quad [\text{W}]$$

with:

- P_{Hy} - Hysteresis losses
- P_{Wk} - Eddy current losses classic
- P_{Wa} - Eddy current losses anomalous
- f - Frequency
- s - lamella thickness
- $\hat{B}$ - Magnetic flux density (peak value)
- ρ_{Fe} - Density
- σ_{Fe} - Electrical conductivity
- d - Magnetic core thickness
- k - Coefficients

Non-grain orientated electrical steel is isotropic and therefore suitable for rotating electric drives.



Graphic: Leaflet 401, SIZ

Classification

Not grain orientated

Grain orientated

Magnetic properties

Isotropic

Anisotropic

Microstructure

Random polycrystalline structure with
20-200 µm crystals

Uniform orientation of crystals in rolling
direction (Goss texture)

Grains mm - cm range

Application

Electrical drives, generators, contactors,
relays, small transformers

High-power transformers

Use of non-grain orientated electrical steel for electric drives,
as there is no preferred direction of magnetization

The labeling system for steels according to DIN EN 10027-1 and DIN EN 10027-2 can be used to clearly classify electrical steel.

Higher Alloy content ↑

Steel grade		Nominal thickness [mm]	Remagnetization loss at 50 Hz and 1.5 T [W/kg]	Min. magnetic polarisation [T] in the alternating field at a magnetic field strength of [A/m]			Anisotropy of the remagnetisation loss at 50 Hz and 1.5 T max [%]	Stacking factor min.	Bending number min.	Agreed density values [kg/dm ³]
Short name	Number			2.500	5.000	10.000				
M210-35A	1.0802	0,35	2,10	1,49	1,60	1,70	±17	0,95	2	7,60
M235-35A	1.0890		2,35	1,49	1,60	1,70	±17		2	7,60
M250-35A	1.0800		2,50	1,49	1,60	1,70	±17		2	7,60
M270-35A	1.0801		2,70	1,49	1,60	1,70	±17		2	7,65
M300-35A	1.0803		3,00	1,49	1,60	1,70	±17		3	7,65

Example standard designation M700-50A according to DIN EN 10106

- M → FeSi Electrical sheets
- 700 → max. magnetization losses 7.00 W/kg (at 50 Hz and 1.5 T)
- 50 → Nominal thickness = 0.50 mm
- A → Not grain orientated

Material number for example M700-50A according to DIN EN 10027-2: 1.0815

Graphic: DIN EN 10106

Designation	Nominal thickness [mm]	Maximum value of the magnetization loss at [W/kg]		Stacking factor min.	Bending number min.
Short name		1,0 T	Frequency [Hz]		
NO 5	0,05	45	1.000	0,88	2
NO 10	0,10	13	400	0,91	2
NO 15	0,15	14	400	0,92	2
NO 20	0,20	15 ^a	400	0,93 ^a	2

^a The value was determined on test strips without coating

Graphic: DIN EN 10303

Thin electrical sheet for use at higher frequencies in accordance with DIN EN 10303

- Specially developed for applications at higher frequencies, e.g. in the automotive industry
- Low losses at high frequencies

Due to the rolling process, the thickness deviation of electrical steel strip is greatest at the edges.

Manufacturing-related influences on the geometric properties:

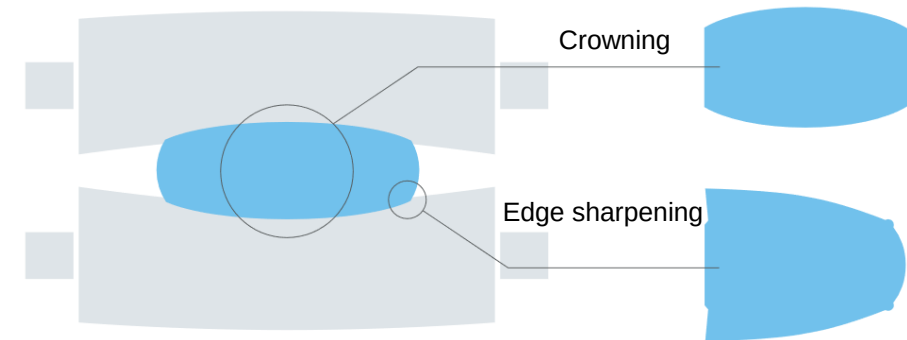
- Thickness tolerances are not uniform across the width
- Sheet metal strips can be clustered with optimization of cut layout
- The sheets with the least thickness variation are in the center.

Maximum geometric deviations:

- Thickness deviations with a nominal thickness of 0.35 mm: $\pm 8\%$
- For the nominal width of 1,250 mm, the limit dimension is 0/+1.5 mm
- The limit dimension of the length is 0/+0.5 mm
- The maximum value of edge curvature at 1 m measurement length is
 - 0.5 mm for nominal widths $l > 150$ mm
 - 1.0 mm for nominal widths $30 \text{ mm} < l \leq 150$ mm

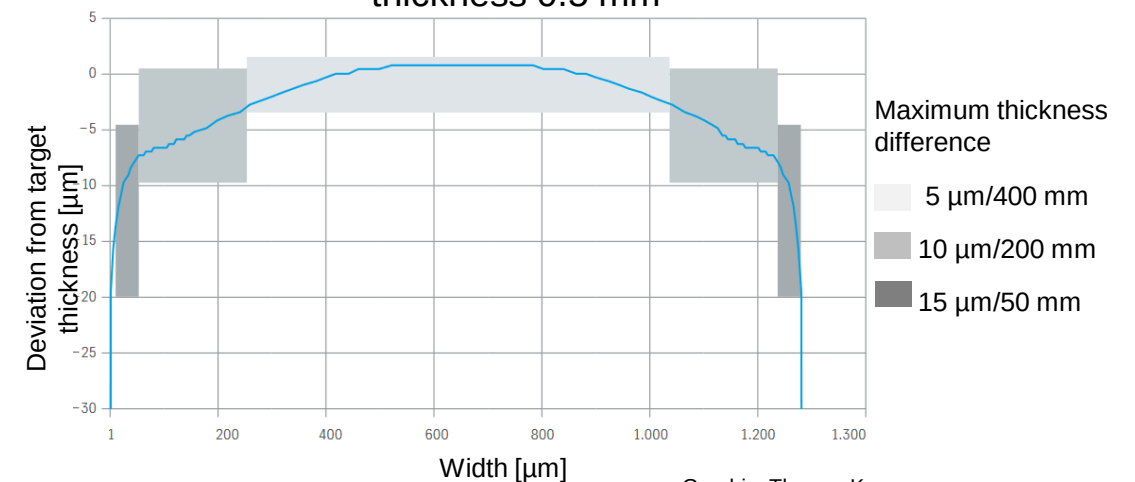
Source: DIN EN 10106

Working roll deflection and embedding of the rolled material



Graphic: ThyssenKrupp

Optimised cutting division of material with nominal thickness 0.5 mm



Graphic: ThyssenKrupp

The saturation flux density is an important characteristic value for soft magnetic materials and indicates the maximum possible magnetization of a material.

Magnetic flux density B :

Property of a magnetic field:

$$\vec{B} = \mu \cdot \vec{H}$$

Magnetic polarization J :

Part of the flux density caused by the magnetic material:

$$\vec{J} = \vec{B} - \mu_0 \cdot \vec{H}$$

Permeability μ :

Magnetizability of the material in an external magnetic field

$$\mu = \mu_0 \cdot \mu_r$$

Saturation flux density B_s :

Maximum possible magnetization of a material

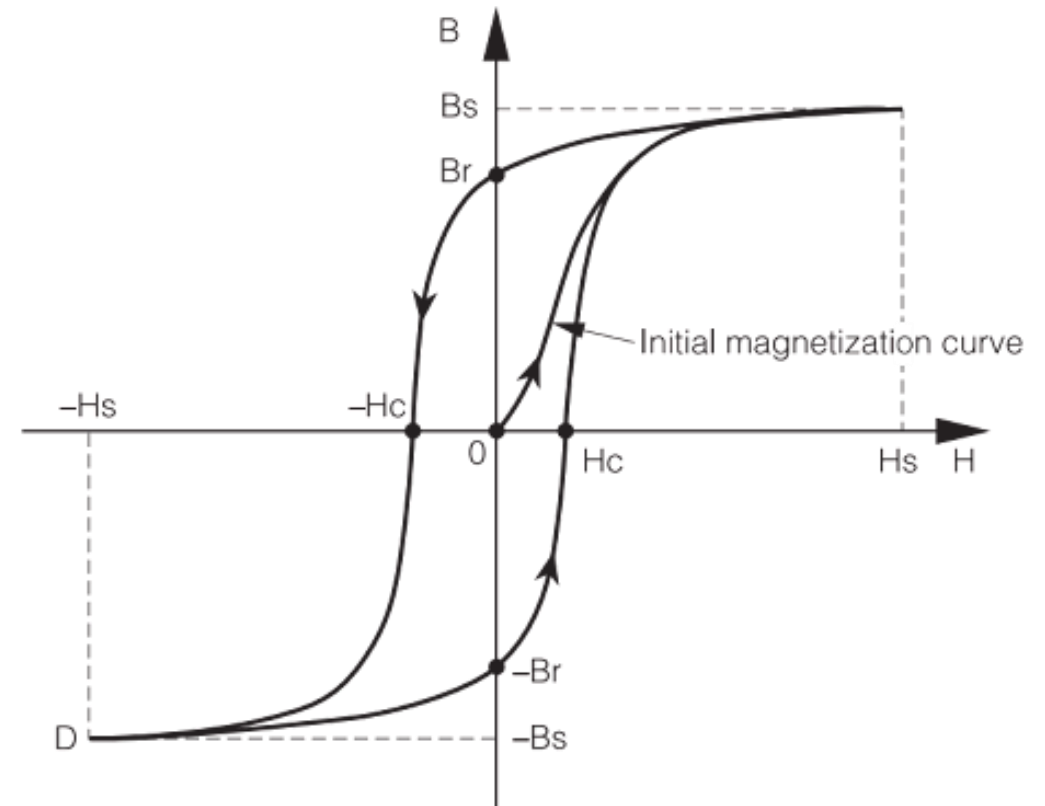
Remanence flux density B_R :

Magnetization that remains after removal of the external field

Coercive field strength H_C :

Field strength for complete demagnetization of a material

Hysteresis loop of a soft magnetic material



Graphic: owenduffy.net

The magnetization curve provides information about the magnetic behavior of the material during the magnetization process.

Phases of magnetization in the alternating field:

1: Non-magnetized material

1-2: As the field strength H increases, **domains** with an orientation parallel to the field lines of the external magnetic field **grow**:

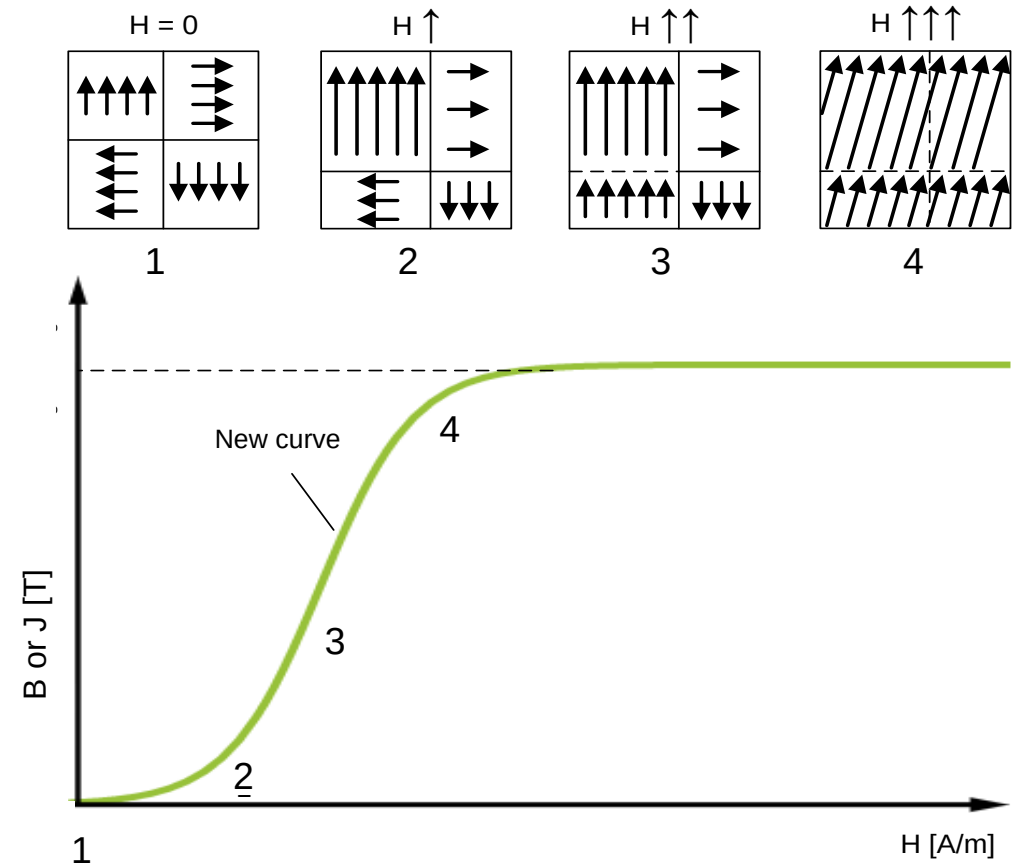
Reversible processes (alignment of the domains (Weiss districts) in the H -field without Bloch wall displacement)

2-3: Irreversible processes in the form of Bloch wall shifts: Enlargement of the domains up to defects → Barkhausen jumps

3-4: Alignment of the domains that are not yet aligned by **rotating the elementary magnetic dipoles**

=> All domains are aligned → Saturation polarisation is reached

New curve for magnetization process in ferromagnetic material



Graphic: based on Callister et al. 2013

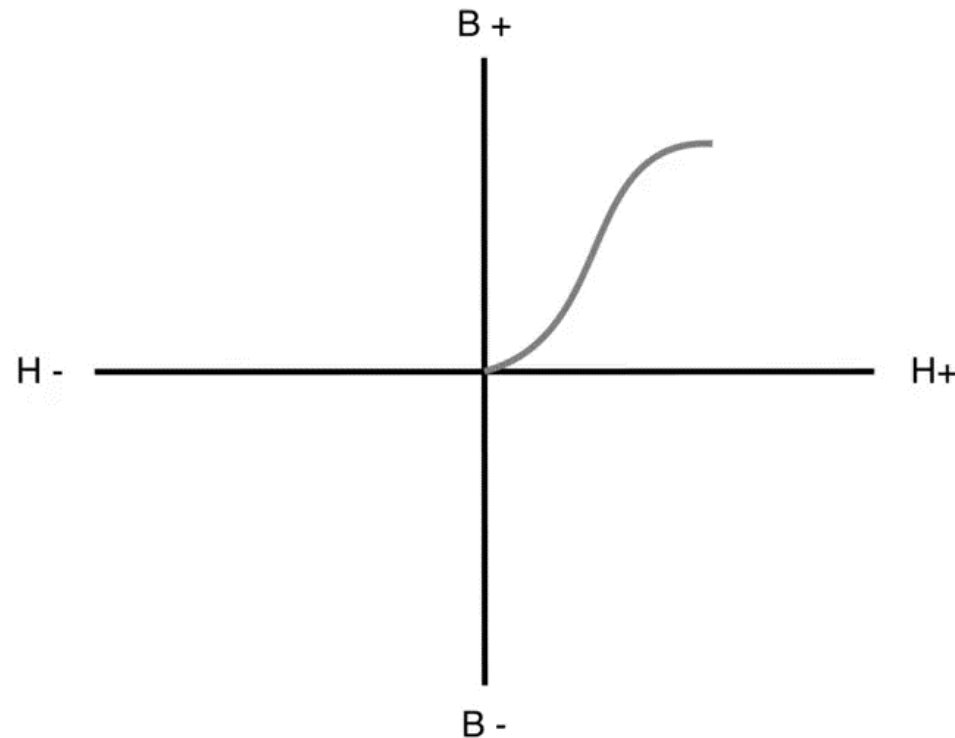
The hysteresis loop shows the relationship between magnetic flux density and magnetic field strength (magnetizing force) of a ferromagnetic material during cyclic magnetization.

Theory

The Hysteresis loop explained

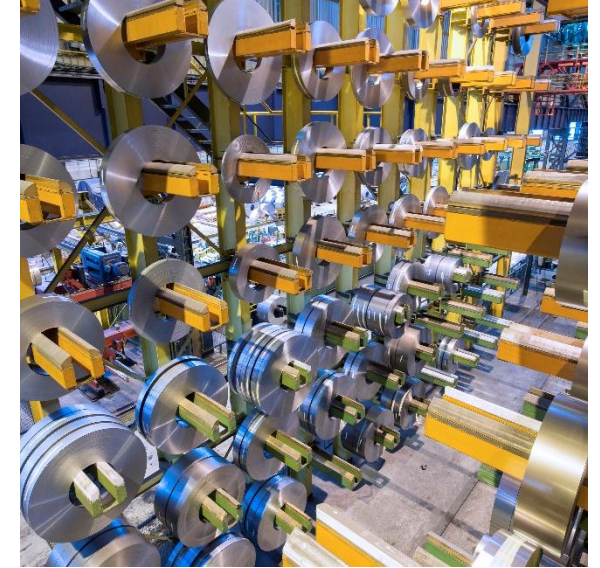
Video:
The Electric Academy

<https://www.youtube.com/watch?v=ZyVSMfSgOq0>



Lecture unit 2 provides an overview of the classification and manufacturing of soft magnetic materials.

- Overview and classification of soft magnetic materials (especially electrical steel)
- **Production of electrical steel**
- Insulating coatings for electrical sheets
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- Alternative soft magnetic core materials

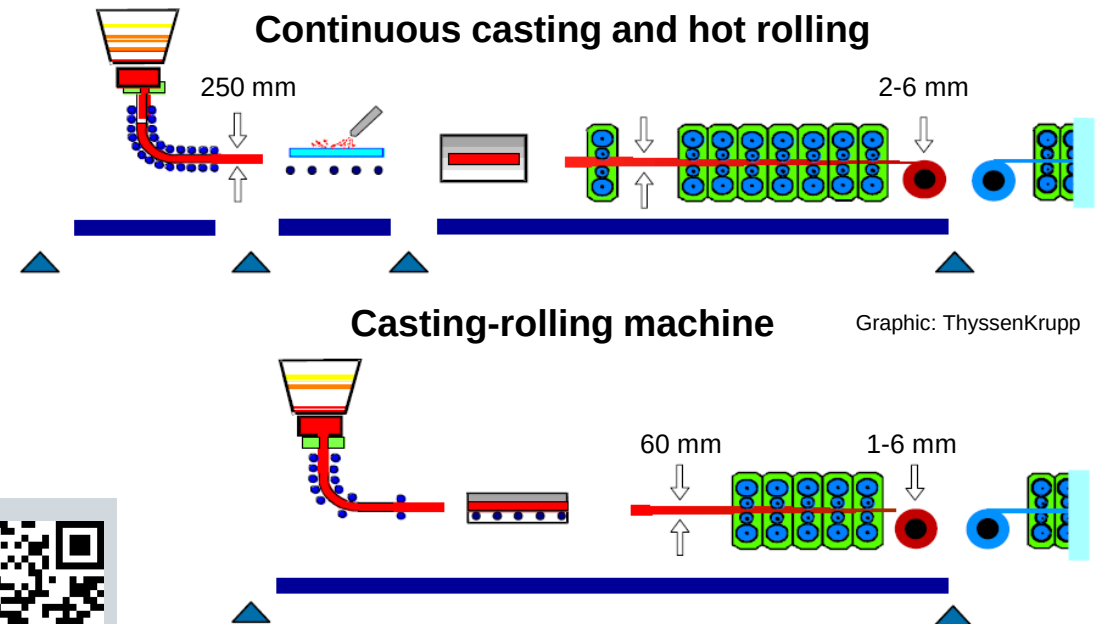


Graphic: ThyssenKrupp

The specific properties of the material are determined during the production process.

The following parameters are checked and adjusted during production:

- Width
- Geometry
- Grain size
- Thickness
- Excretion
- Texture
- ...



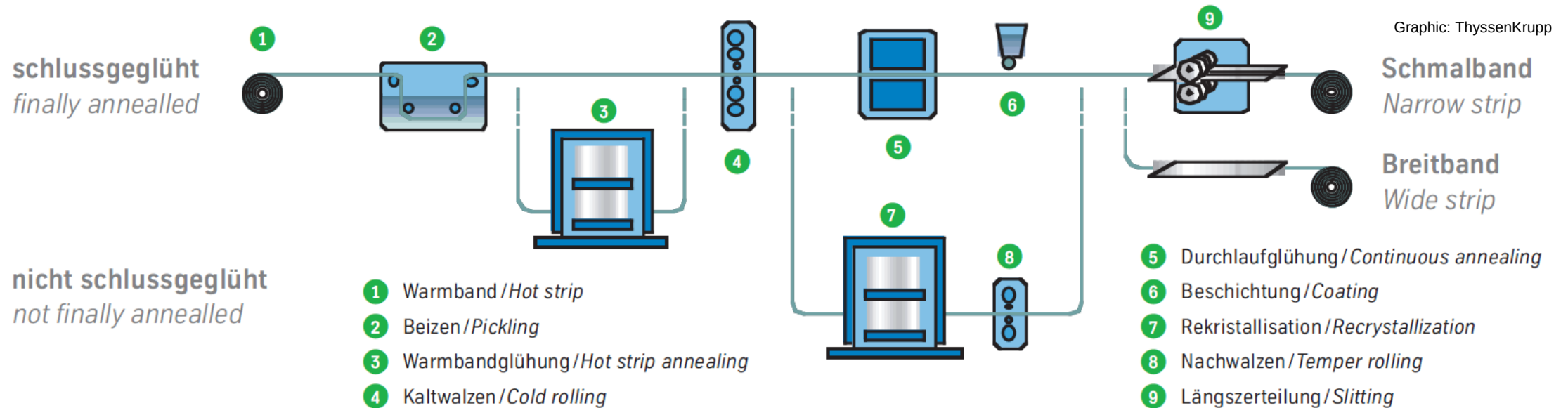
Video:
Reibus

<https://www.youtube.com/watch?v=a7cEJpH9UiQ>



Non-grain-oriented electrical steel is divided into fully-finished and semi-finished electrical steel.

Difference in the process chain in NKO electrical steel production between semi-finished and finished products

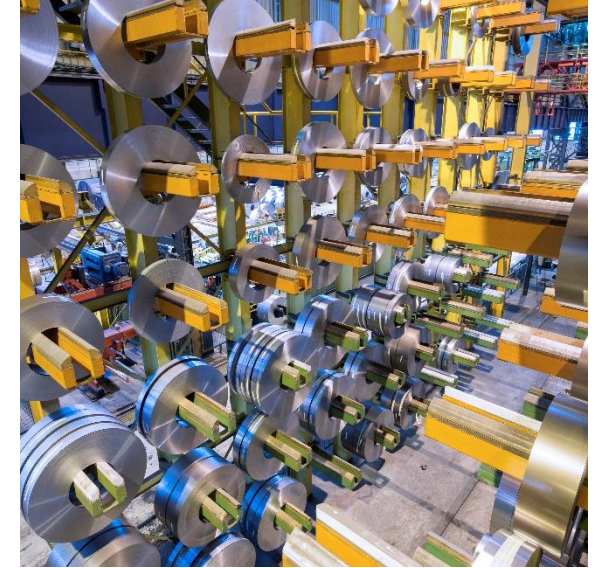


There are the following differences between semi-finished and fully-finished electrical steel grades

- The magnetic properties are adjusted during annealing: much easier to adjust with fully-finished electrical steel strip
- Semi-finished production method preferably used in USA
- Additional systems required for annealing in the processing plant for semi-finished material
- Possibility of subsequent elimination of production influences with semi-finished material

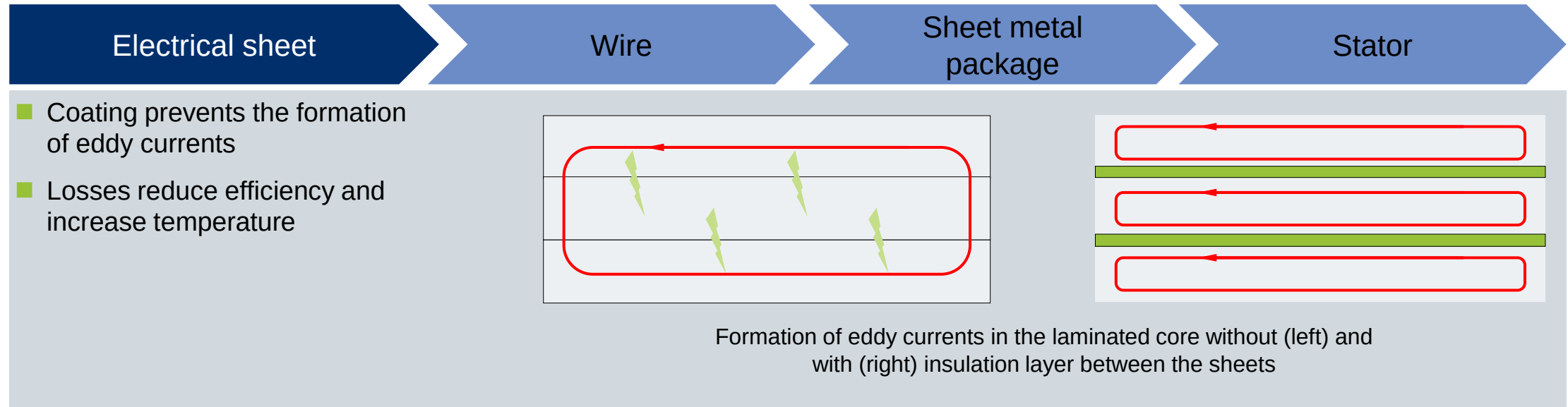
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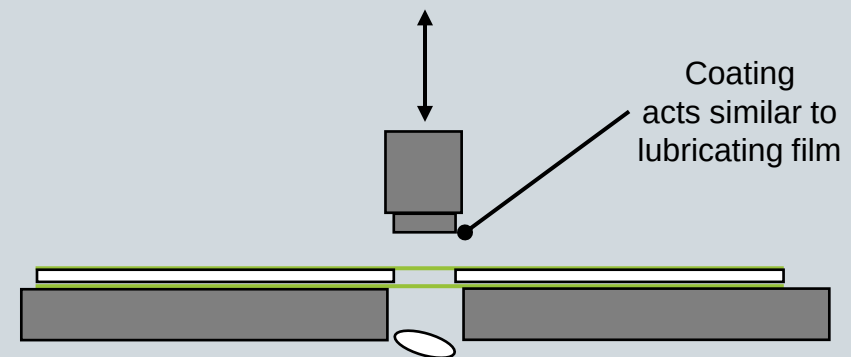


Graphic: ThyssenKrupp

In addition to electrical insulation, the insulation of the sheet metal package also provides corrosion protection.



- Pre-applied insulating layer acts as a lubricating film during punching
- Improves cutting properties of otherwise brittle electrical sheet metal
- Increases tool life
- Protects against short circuits caused by burrs



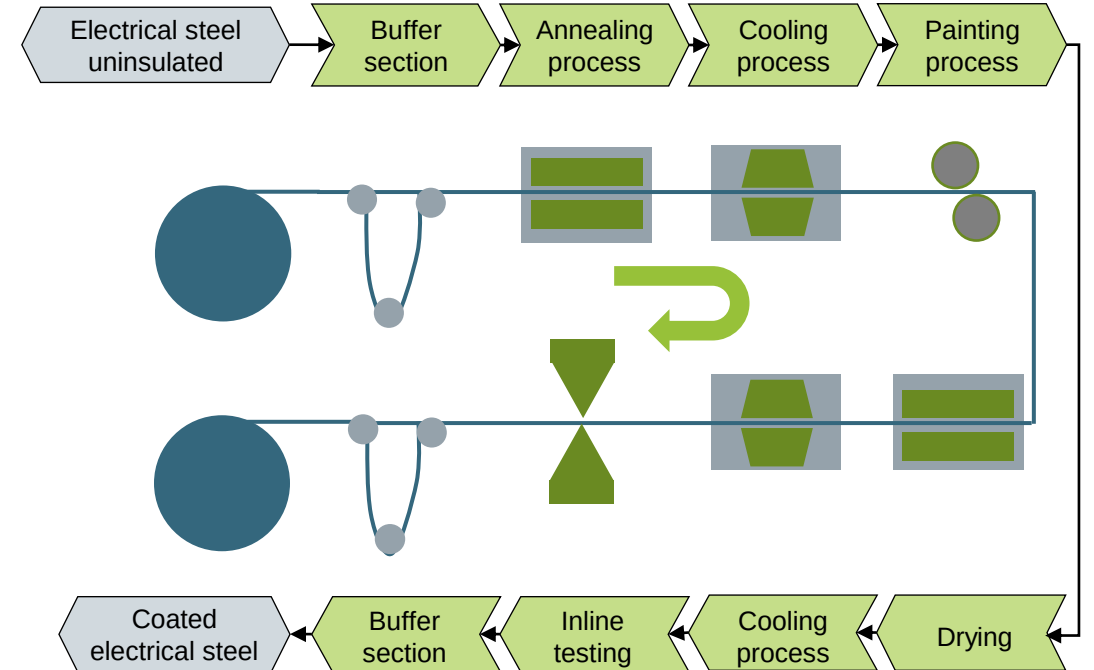
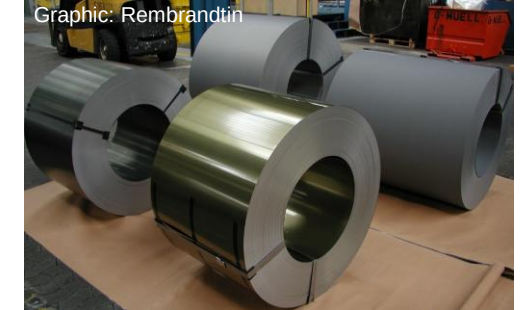
To avoid additional eddy current losses in lamination stacks the single sheets are insulated on one or both sides.

Requirements for electrical steel insulation systems:

- Improving the cutting properties
- No spalling and no cracking when bent
- Low-porosity, homogeneous and thin ($< 5 \mu\text{m}$ on one side)
- Pressure-resistant up to $1,500 \text{ N/cm}^2$ at 150°C
- Environmentally friendly and durable
- Possibly weldable and annealing resistant

Application to the sheet metal coil

- Application at the sheet metal manufacturer using paint rollers
- Re-insulation at the sheet metal fabricator if required
- Inline measurements of the relevant parameters
- Additional coating with bonding varnish systems if necessary



Graphic: based on Axalta

The insulation reduces eddy current losses during the operation of electrical machines and thus increase their efficiency.

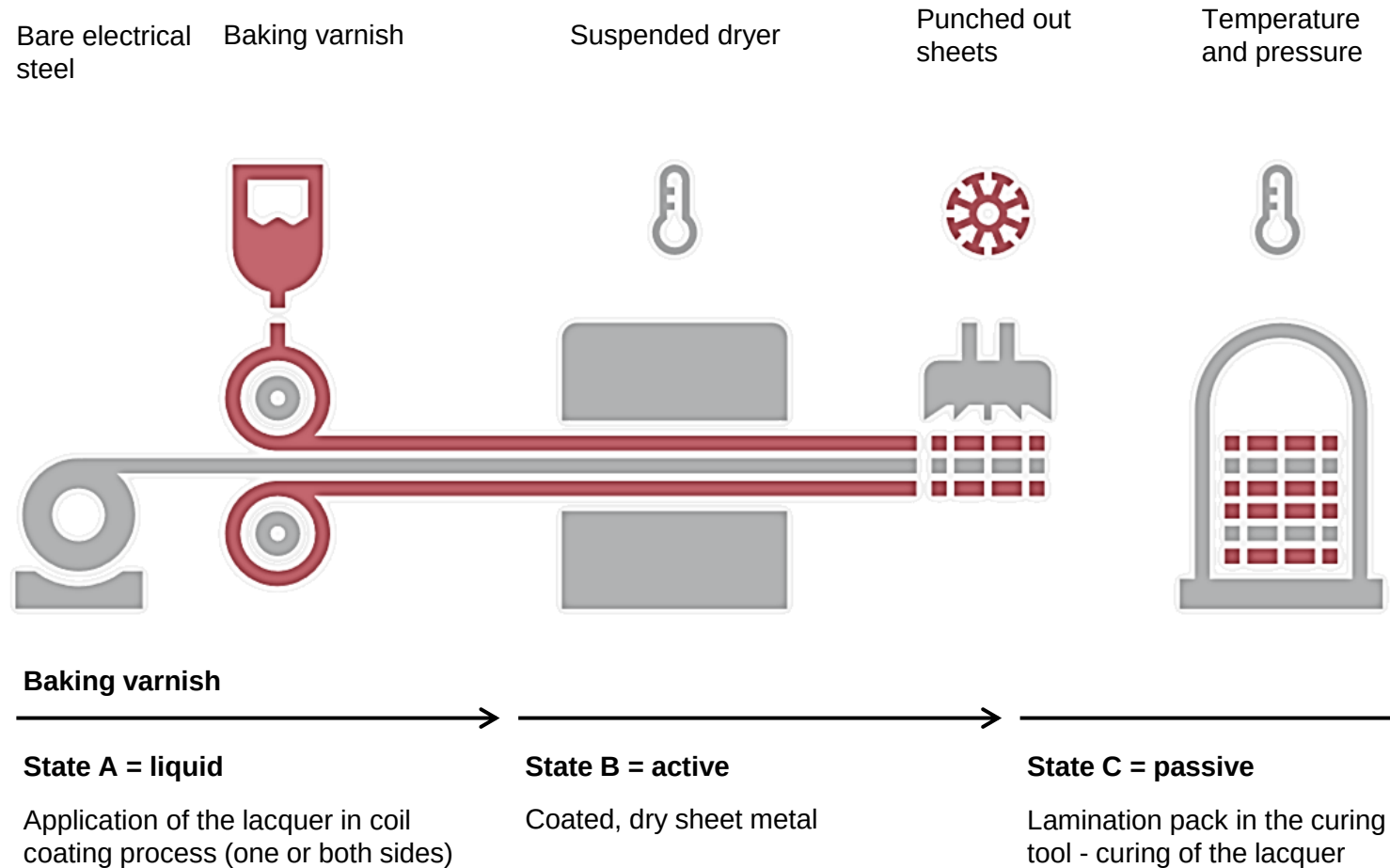
Insulation class	Layer thickness per side [μm]	Permanent resistance under air [°C]	Annealing resistance under inert gas [°C]	Insulation resistance [Ω]	Properties	Application	Composition
C-3	0.5 - 6	180	-	10 - 200	Limited weldability, small motors	Finally annealed, non-grain orientated electrical steel strip	Unfilled organic-based lacquers, increased punchability
C-5	0.5 - 5	210	600	5 - 200	Good weldability, suitable for stress-relief annealing	Non-grain orientated and grain orientated electrical steel	Inorganic coating with ceramic fillers, increased punchability
C-6	2 - 7	180	500	> 10000	Pressure-resistant, highest insulation properties	Finally annealed, non-grain orientated electrical steel strip	Organic coating with inorganic fillers, increased punchability

Sources: DIN EN 10342

For the subsequent bonding of electrical steel packs with baking varnish an additional layer of lacquer is applied to the electrical steel strip.

Properties of baking varnish insulation:

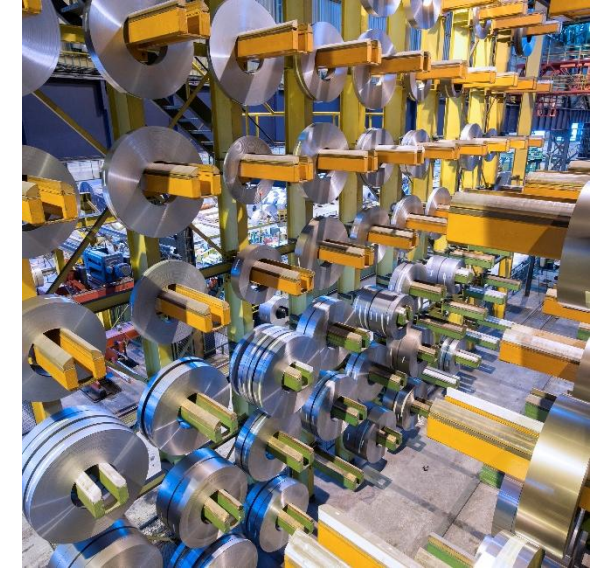
- Solvent-based or water-dilutable
- Temperature for bonding: 130 - 250 °C
- Shelf life in condition B: approx. 12 months
- Activation during bonding by: Pressure, temperature and time
- Layer thickness: 3 - 8 µm



Graphic: CDW

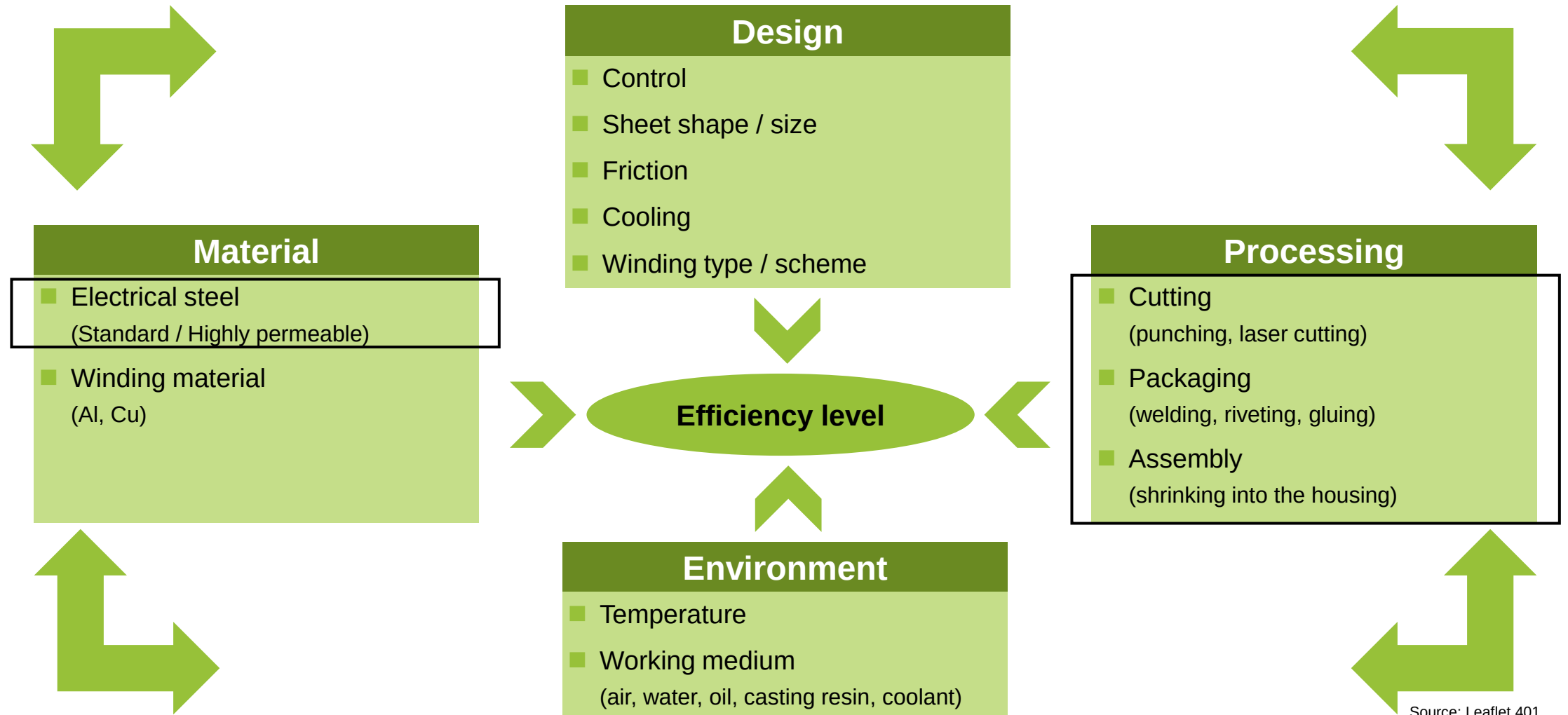
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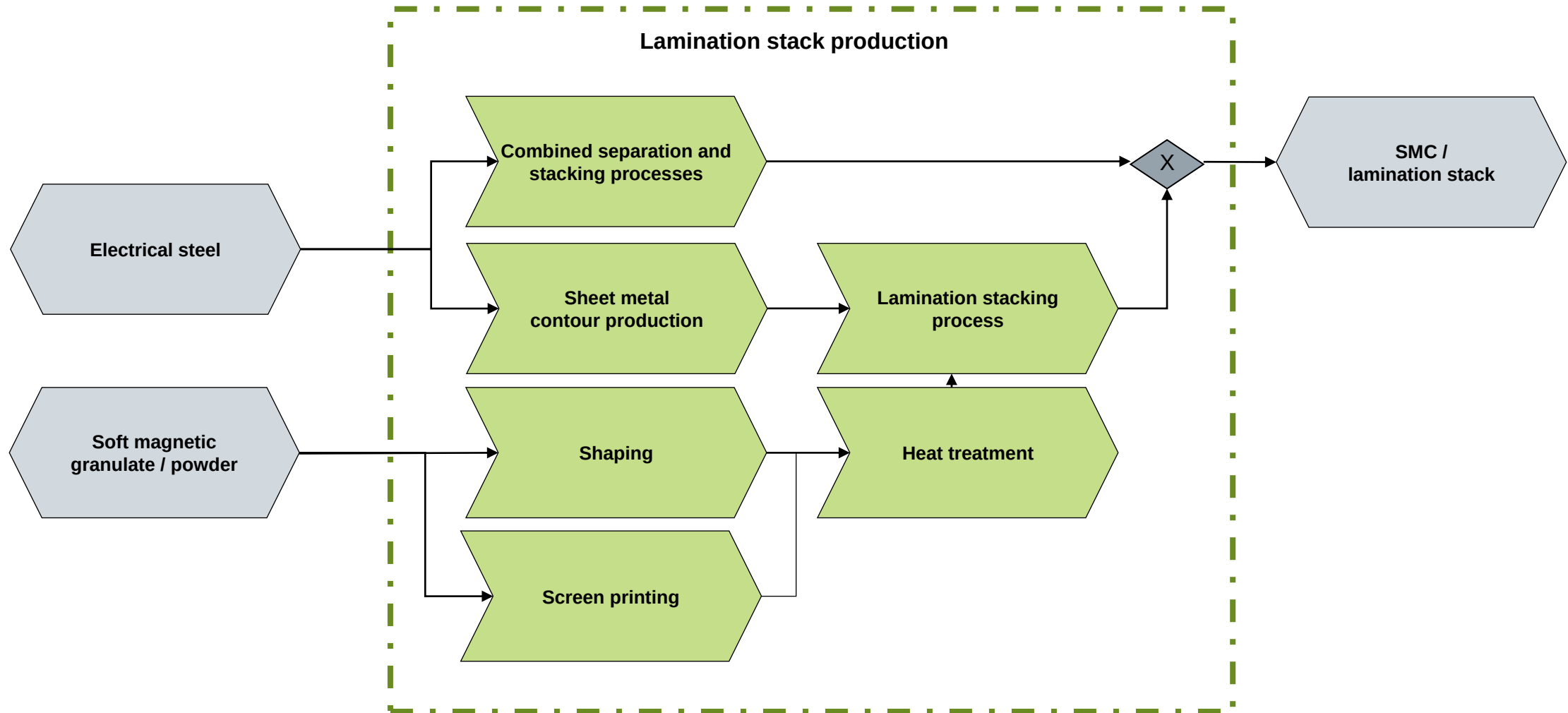


Graphic: ThyssenKrupp

The processing methods significantly influence the efficiency in electromagnetic systems.



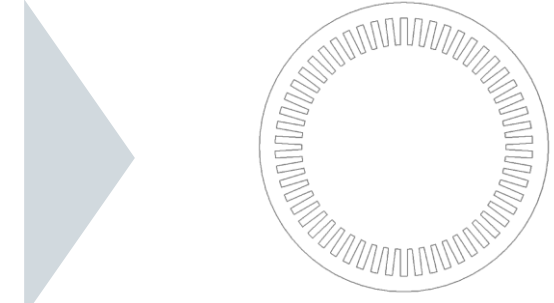
The soft magnetic core consists either of laminated electrical steel sheets or of sintered powder composite materials.



The segmentation of the laminations leads to an increased sheet utilization and is suitable for single-tooth coils, for example.

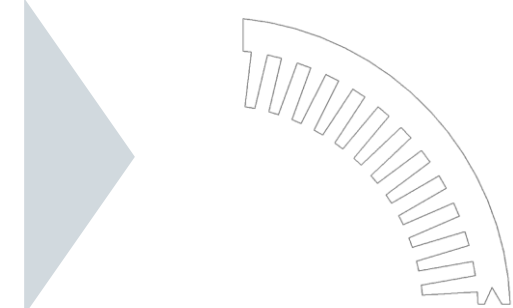
Full cut:

- Very widespread
- Low joining and assembly effort
- Rotor cut integrated to increase sheet utilization



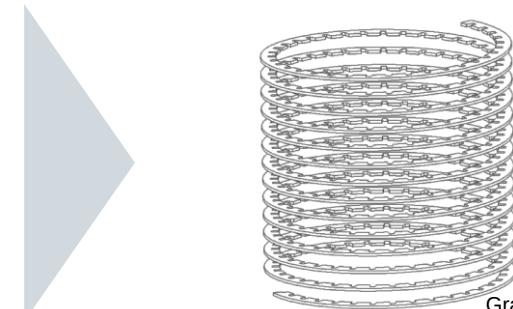
Segmentation:

- Various forms of segmentation possible
- Increased sheet utilization, especially with different rotor and stator materials
- Increased copper fill level due to freely accessible winding space
- Additional assembly step required



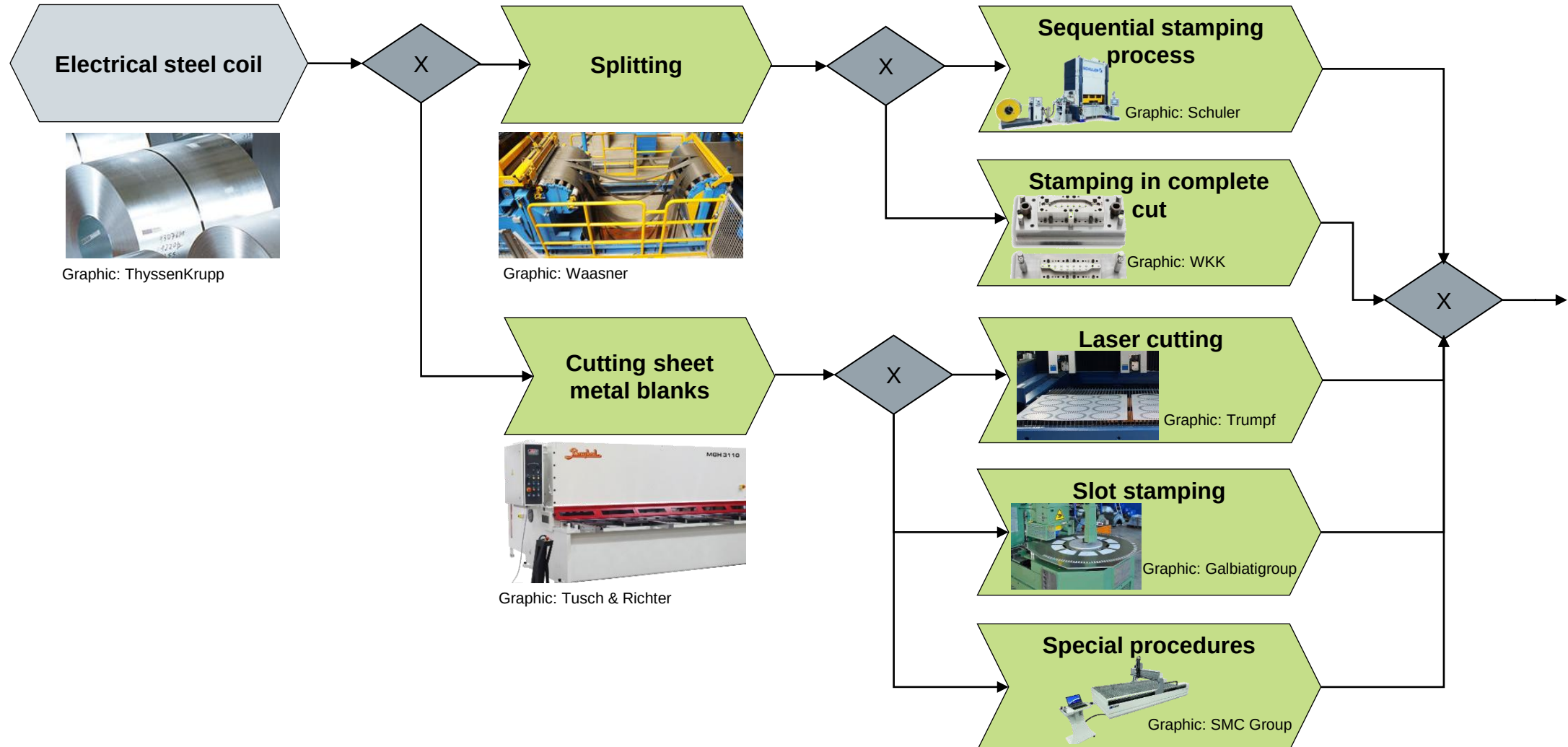
Spiral-shaped stacks ("slinky technology"):

- Endlessly assembled stacks
- Production by forming operation
- Use with electrical machines with lower requirements



Graphics: FAPS

The selection of manufacturing processes for the production of lamination stacks depends on the quantity and the requirements.



The selection of manufacturing processes for the production of lamination stacks depends on the quantity and the requirements.

Sequential stamping process:

- Suitable for high quantities
- Punch out the final shape in cycles
- Fully automated process, optionally with stacking

Stamping in complete cut:

- Suitable for small to medium quantities
- Punch out the complete final contour with one stroke
- Removal and stacking of stamped parts mostly by hand

Slot stamping:

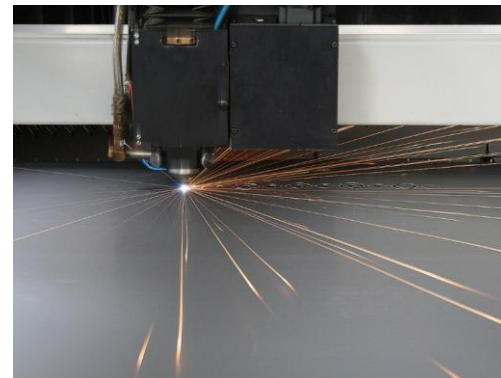
- Suitable for smaller quantities
- Punch out the individual slots in cycles
- Manually operated or fully automated

Laser cutting:

- Prototype production, small series
- Cutting out the contour with a laser beam
- Manual loading and removal of trays



Graphic: Schuler



Graphic: Grau Stanzwerk



Graphic: Schuler

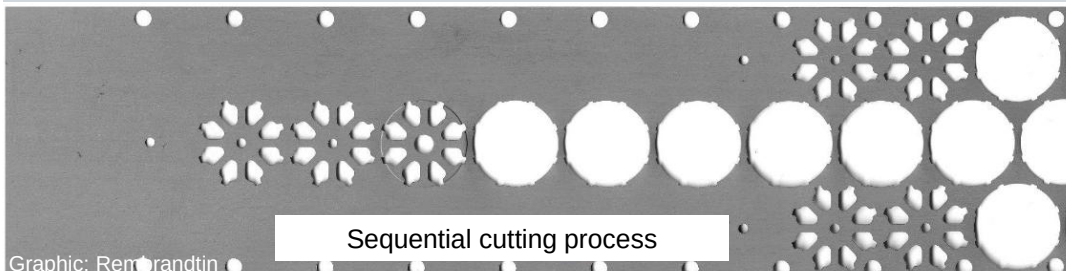
Punching is particularly suitable for separating electrical steel laminations in mass production.

Factors that influence the dimensional accuracy of stamped parts

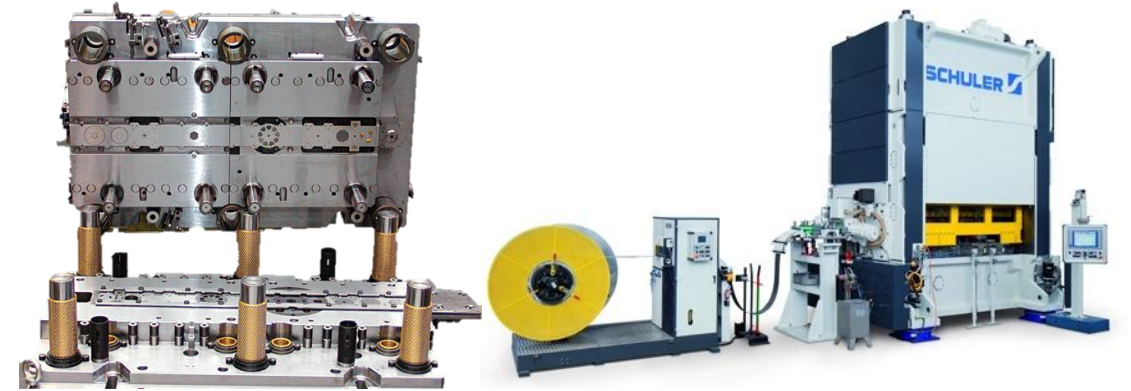
- Precision of the mold and the press
- Geometry of the parts
- Surface of the electrical steel strip

Properties of the stamping process

- Optimal cutting gap at approx. 10% of the sheet thickness
- High productivity with 150 to 400 strokes per minute
- Improved punchability via insulation layer or lubricant
 - Stamping burr increases with tool wear
 - Deterioration of the stacking properties of the lamination stack, possibly causing short circuits between sheets



Graphic: Renrandtin

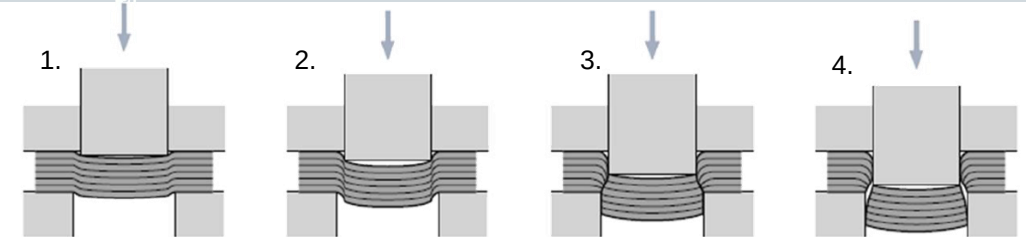


Graphic: Sitem

Graphic: Schuler

The four phases of the stamping process

1. Sheet metal deformation due to contact with the stamp
2. Cutting the sheet metal
3. Sheet breakage at the cutting contour due to the high internal stresses
4. Ejection of the stamping slug



Graphic: Fascination sheet metal, TRUMPF

Slot stamping of electrical sheets is particularly suitable for small quantities and large diameters.

Full stamping process

- Can be used for large and complex components
- Lower stroke rates than for sequential stamping
- No integrated stacking possible
- Processing of band material possible
- Production of the complete contour

Manual or fully automated slot stamping

- Can be used for large components
- Low cycle times due to the processing of individual slots
- No integrated stacking possible
- Automated stamping and stacking with fully automated slot stamping
- Production of one slot per stroke

Sequential stamping

- Cycle-based stamping of smaller components in a multi-stage process
- Particularly material-saving
- Integrated stacking possible
- Processing of coil material in high-speed presses with high stroke rates
- Production of partial contours per stroke

Video:
Kienle & Spiess

<https://www.youtube.com/watch?v=cnSdf2Cjsv0>



Video: Kienle & Spiess

Video:
Hagel
Automation

<https://www.youtube.com/watch?v=QZ278q2dA6U>



Video: Hagel Automation

Video:
Kienle & Spiess

<https://www.youtube.com/watch?v=nFwFrv9-MQY>



Video: Kienle & Spiess

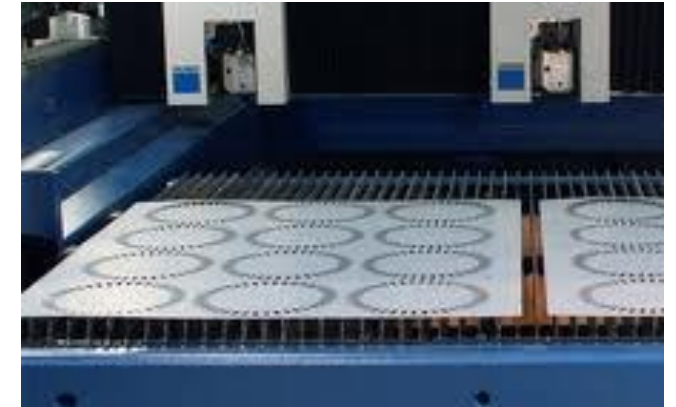
Laser cutting of electrical steel is used in particular in prototype production and for large drives.

Properties of laser cutting:

- Laser cutting enables the realization of flexible cutting geometries
- Laser combines functions that are distributed across several machines in stamping processes
- Fully automatic and highly dynamic machining process
- Use in small series and prototype construction
- Laser cutting parameters must be adapted to the corresponding application and material

Two possible beam source types:

- CO₂ - Laser:
 - To date, the standard laser source for sheet metal cutting
 - Power levels above 1 kW
 - High space requirement, complex, heavy machining heads
 - 10 m/min per linear motor
- Solid-state laser:
 - Fiber laser with very good beam quality, good power scalability, compact dimensions
 - Reduced size and weight compared to CO₂ -lasers
 - Several kilowatts of laser power
 - Cutting speeds up to 20 m/min



Graphic: Trumpf



Graphic: Trumpf

The laser cuts the sheets completely contact-free, resulting in low mechanical stress compared to stamping.

Advantages of laser cutting compared to stamping processes:

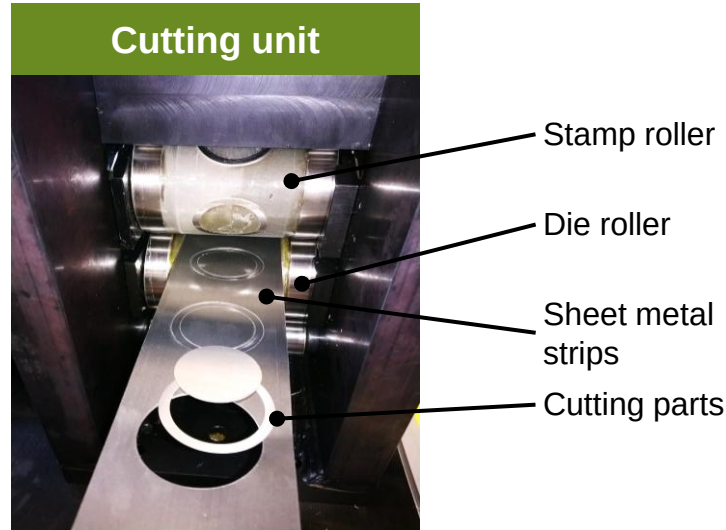
- No cutting tools required
→ Also suitable for small quantities
- Non-contact, force-free
→ No stamping burr
- Processing of very thin or high-alloy sheets possible

Disadvantages of laser cutting compared to stamping processes:

- Heat-affected zone with negative effects on the magnetic properties of the material
- Long cycle times
- Manual loading and unloading



With rotary cutting, the sheets are cut continuously with two rollers, resulting in an even distribution of force.



Advantages compared to stamping process

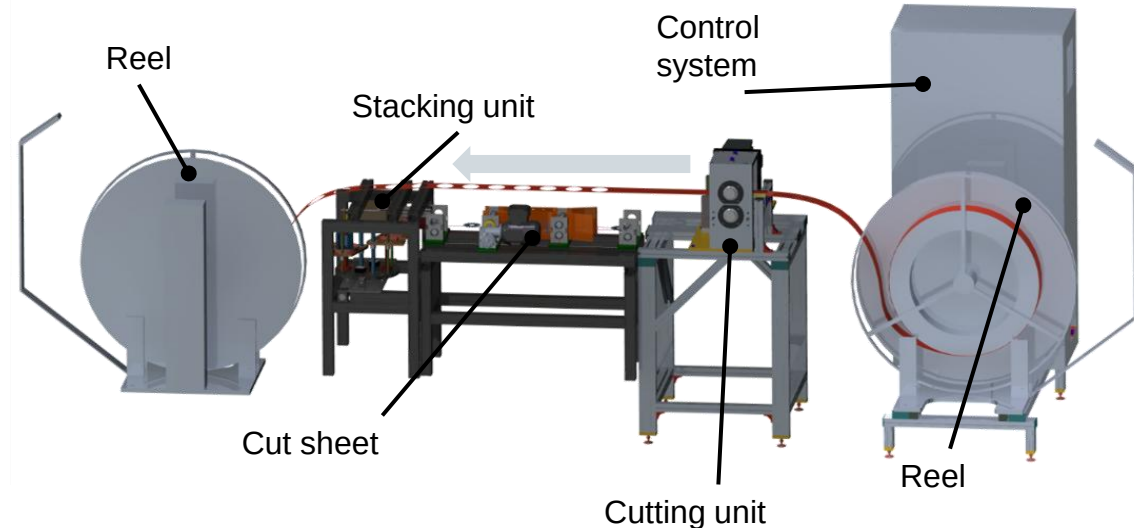
- Continuous process sequence and homogeneous force distribution
- Low moving masses (compared to punching)
- Suitable for very thin sheets
- Lower tool and system costs

Disadvantages compared to stamping processes

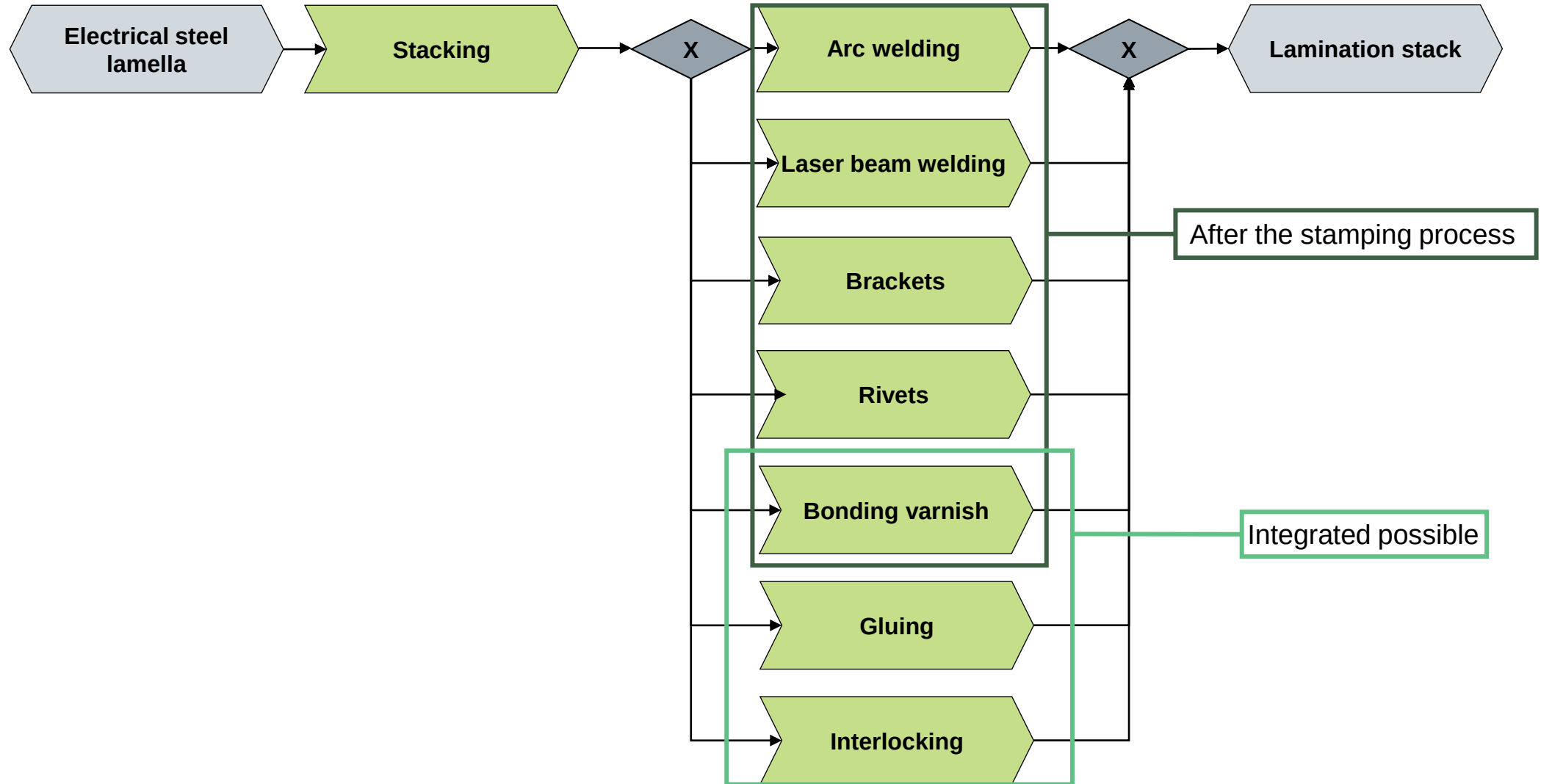
- Technology at the research and development stage
- Tool life and tool wear not final analyzed
- Integrated stacking not yet possible
- Removal of the cut sheets

Properties of the process

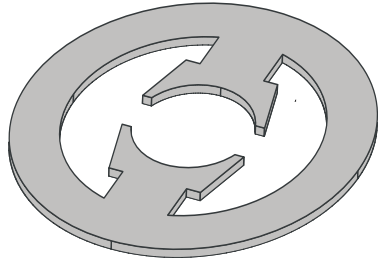
- Sheet metal cutting using two counter-rotating tool rollers
- Punch and die rollers required for cutting electrical steel strip
- Suitable for processing wound material
- Compact system topology and tools



The selection of manufacturing processes for the production of electrical sheet packages depends on the quantity and requirements.



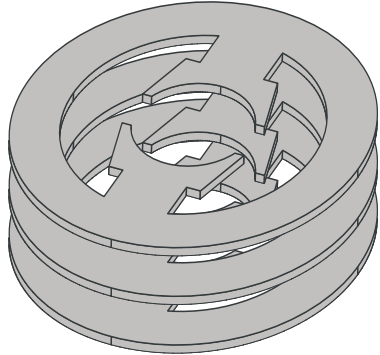
Various challenges must be taken into account when stacking metal sheets.



**Hooking between the
single sheets**



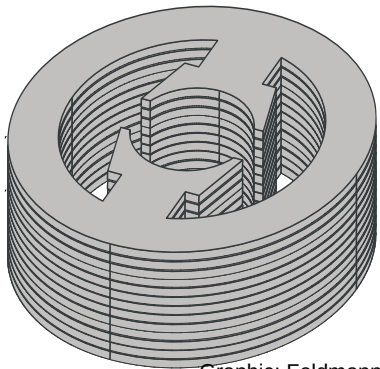
Difficult to separate



Burr formation



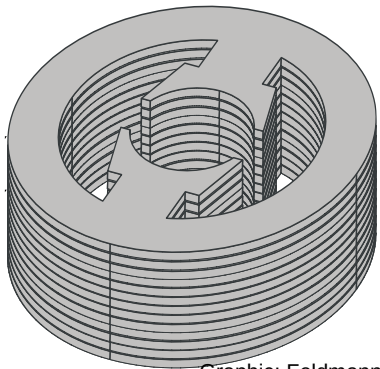
**Faults during separation
and by abrasion**



**Complex
inner contour**



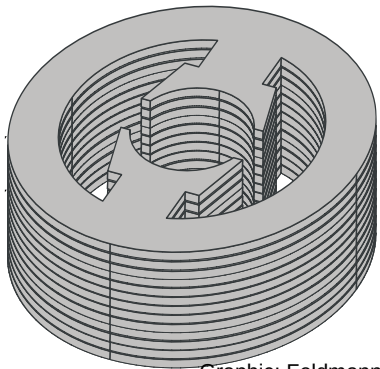
Hook when pushing off



Flexible parts



**Disturbances due to
deformed parts**



**Sheet thickness
tolerance**



**Misalignment of the
lamination stack**

Graphic: Feldmann 2006

Depending on the size, there are specific requirements for the assembly of lamination stacks.

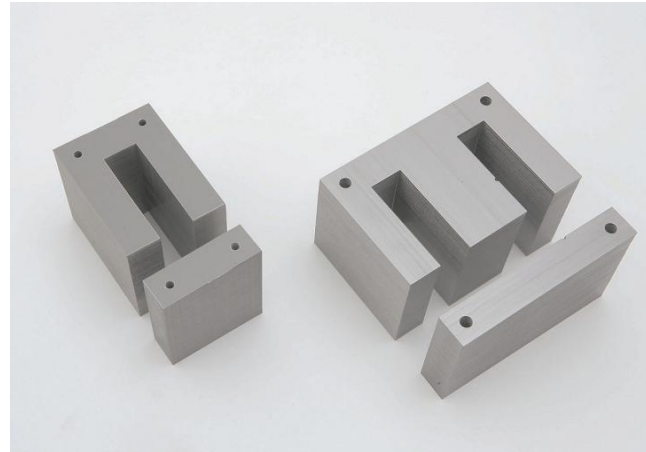
Requirements for the stacking process:

- Compensation of thickness fluctuations
- Compliance with the required tolerances
- With large machines, the stacking process is usually carried out manually
- Achieving a high stacking factor
- No damage to the insulation layer

Electrical machines



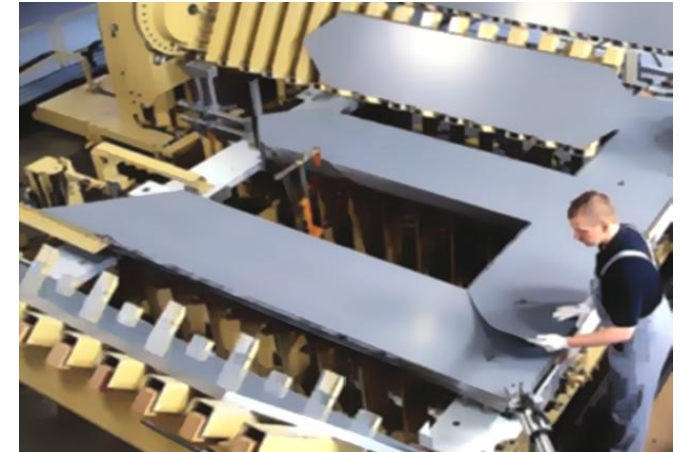
Chokes, solenoids, transformers, etc.



Size

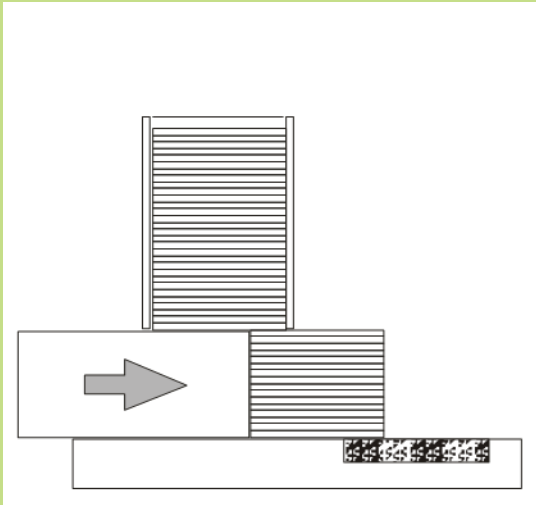


Graphics: Rose, Steel Centre, Stanzwerk



Mechanized stacking of sheet packages can be accomplished using alternative strategies.

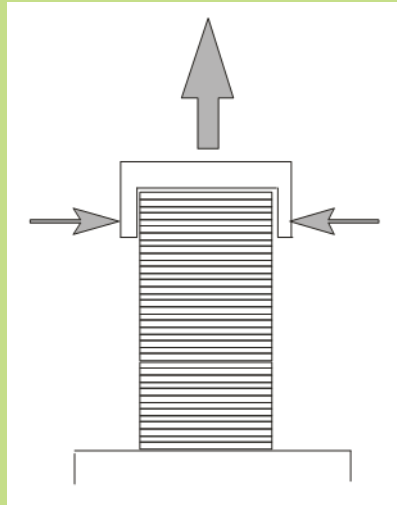
Lateral pushing off the complete sheet stack



Predominant processes: for smaller sheets

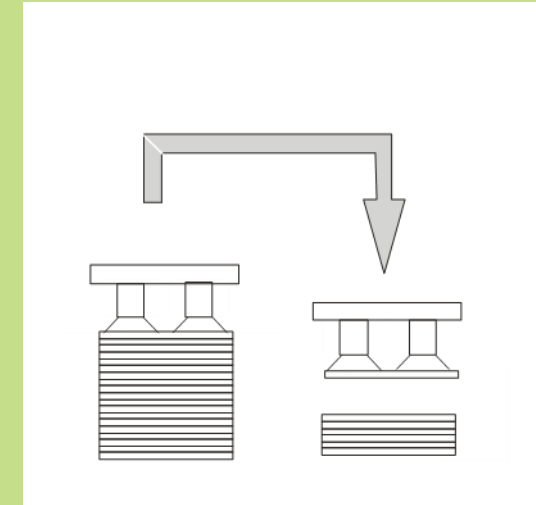
- Simple kinematic structure
- Susceptible to faults with sheet thickness fluctuations and complex internal shapes

Coaxial removal of complete sheet packs



- Secure gripping even with complex internal moulds
- Susceptible to faults with very flexible parts

Stacking individual sheets



For larger sheets

- Safe but complex and slow procedure
- Separation often difficult due to adhesion

Graphics: Feldmann 2006

The individual sheets are brought together using various joining technologies to form the lamination stack.

Common joining methods:

- Hot and cold bonding process or bonding varnish technology (Graphic 1)
Full-surface connection of the individual sheets
- Brackets (Graphic 2)
Attach an additional fastener to the outer contour of the sheet metal package
- Welding (Graphic 3)
Welding of the individual sheet metal lamellas using various processes
- Interlocking (Graphic 4)
Integrated joining process using interspersions in the sheet
- Rivets (Graphic 5)
Holes / recesses at the intended location on the sheets
- Plastic packaging (Fig. 6)
In injection molding machine → Creation of the desired shape

Some joining methods create **stresses** in the material or **short circuits** between the sheets.

→ **Deterioration of the magnetic properties**

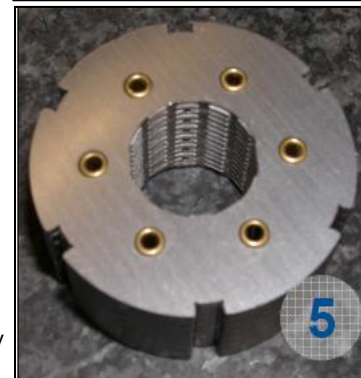
Bonding varnish process



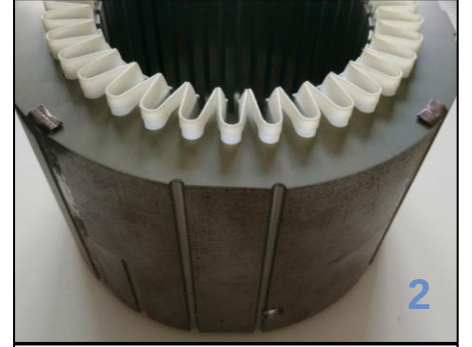
TIG welding



Rivets



Brackets



Interlocking



Plastic packaging



Graphics:
FAPS, LaserCut AG
/ leaflet 401, SiZ,
Stamping technology
Jessen

Welded lamination stacks fulfil high mechanical requirements and can also be used for large stators.

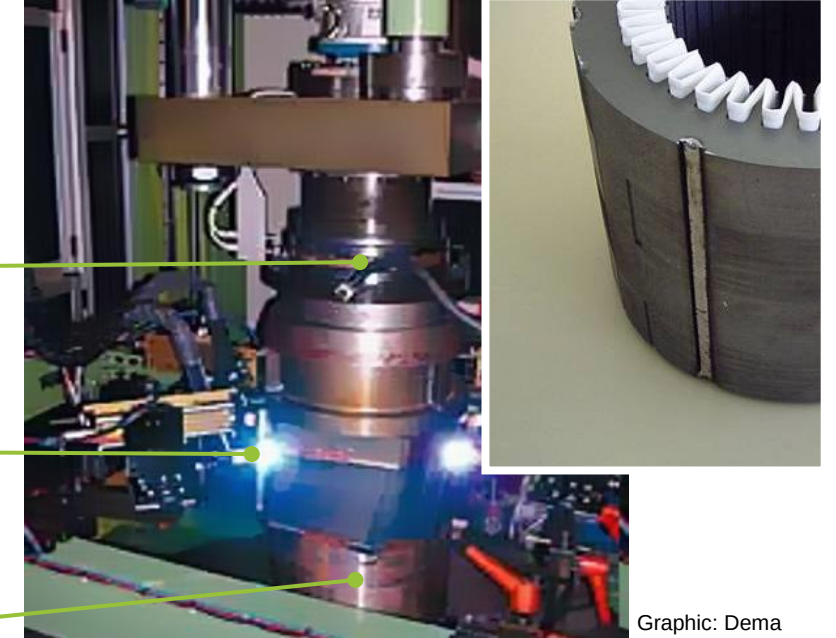
Welding of lamination stacks

- Bonding of individual sheets by heat -> Mechanical strength
- Welding of laminate stack on 2, 4, 6 or 8 seams
- Can be used fully automatically in high volume production or manually in low volume production
- Insertion of welds into recesses to increase geometric tolerances
- Partial weld preparation required

Pressing device

Arc / Flame

Lamination stack



Graphic: Dema

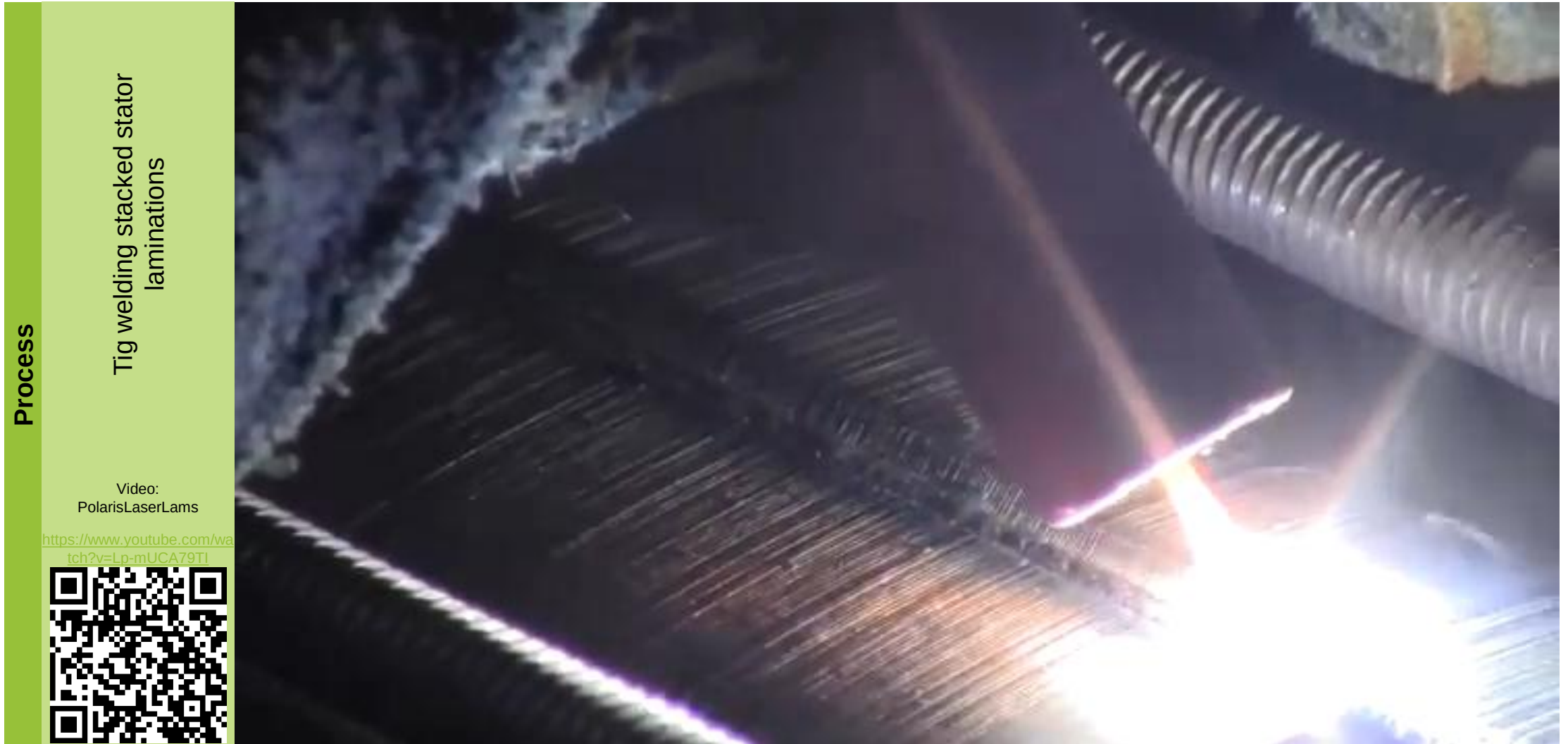
Advantages

- Proven industrial process
- High mechanical strength of the joint
- Processing of large and caseless stators possible
- Chemically and temporally stable joining compound

Disadvantages

- Creation of an electrically conductive connection between the insulated sheets
- Damage to the material structure due to heat-affected zone
- Component distortion possible due to heat input
- No full-surface joint connection

Welded lamination stacks fulfil high mechanical requirements and can also be used for large stators.



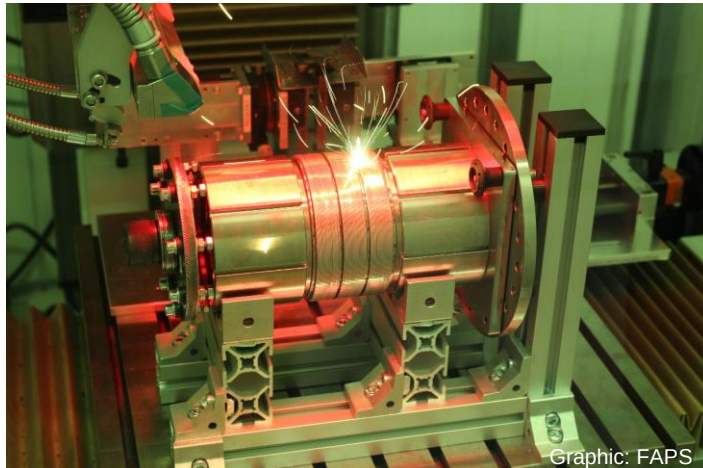
In addition to the classic arc processes, laser and plasma welding processes are also suitable for processing electrical steel laminations.

Laser welding of electrical steel laminations

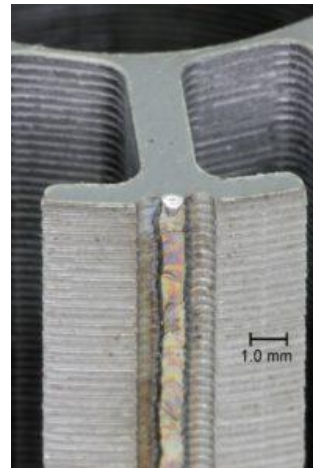
- Low maintenance
- Low delay
- Higher welding speed and higher seam strength
- Weld seam rather deep than wide
- Even thicker insulation and thinner sheets can be welded
- Less susceptible to material fluctuations

Plasma welding of electrical steel laminations

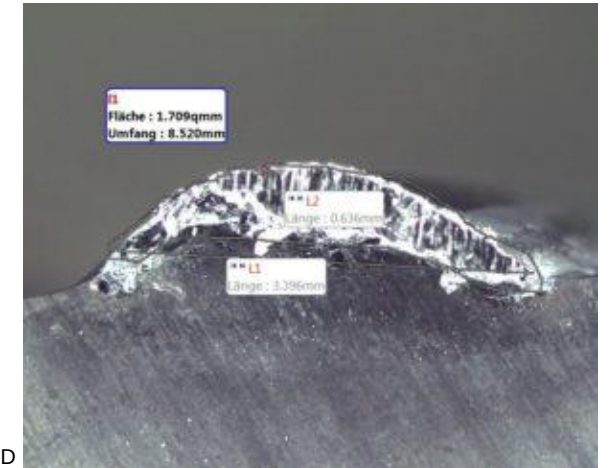
- Minimal warpage and optimum straightness
- High availability
- Reduction of splashes
- Narrow beam focusing
- Control of the process parameters
- Short process times due to the use of multiple burners



Graphic: FAPS



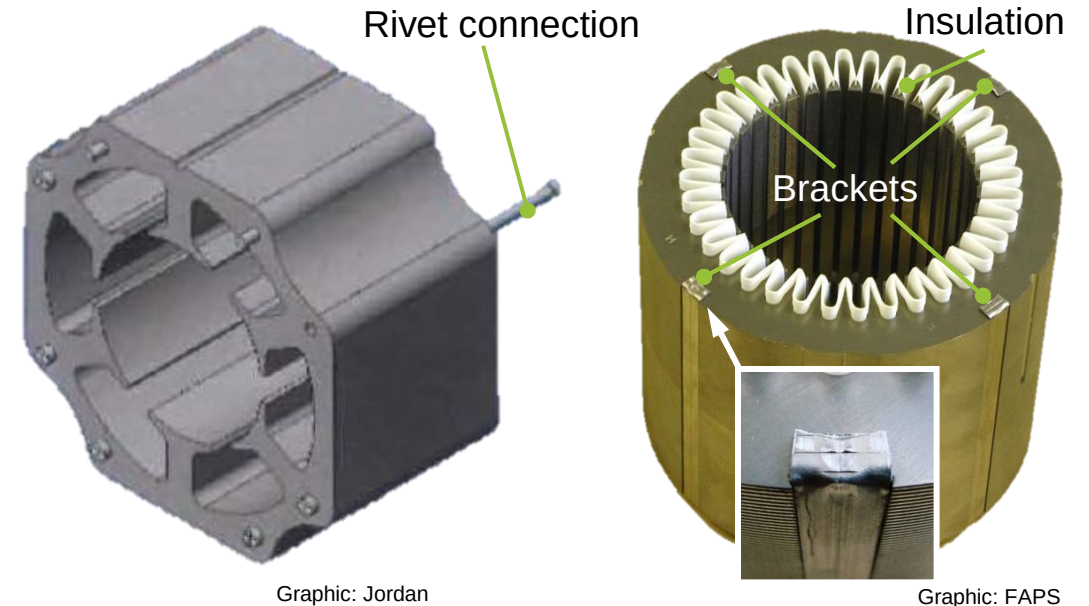
Graphics: SWD



The use of brackets or rivets is cost-effective and quick.

Brackets / clamps and rivets in the packaging process

- The individual sheets are fixed in place with an additional connecting element
- Brackets are applied to the outside of the pressed sheet stack
- Rivets are inserted into additional recesses and fixed in place by forming
- Brackets create a form-fit connection with the groove wall
- Suitable for compact standard motors that are mounted in a housing and have to fulfil low geometric tolerances



Advantages

- Proven industrial process
- No impairment of the material structure
- Can also be used manually for small quantities
- Chemically and temporally stable joining compound

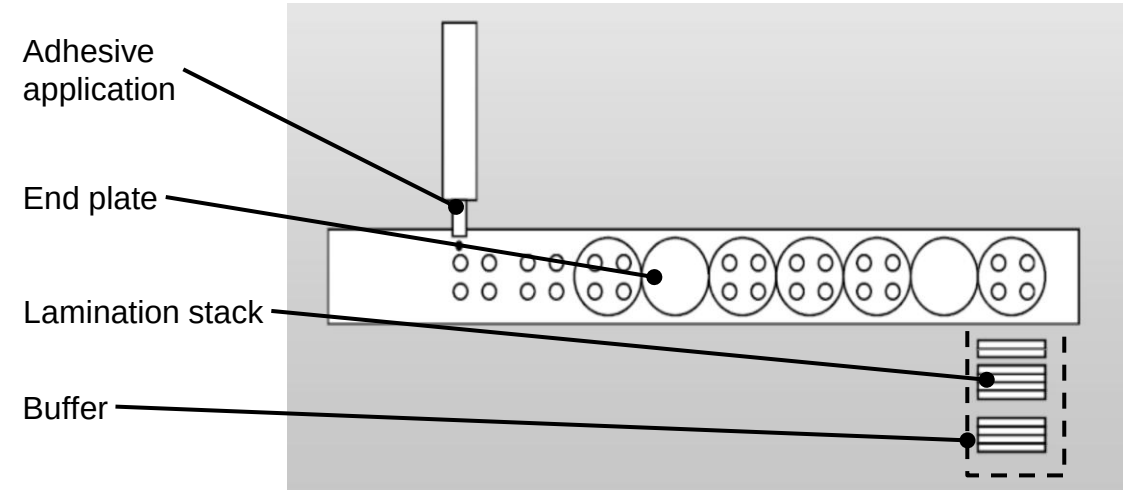
Disadvantages

- Creation of an electrically conductive connection between the insulated sheets
- Low degree of automation and low mechanical strength
- Not suitable for high-quality mass production applications
- No full-surface joint connection

The integration of stacking and joining into the punching process using adhesives shortens the process chain.

Bonding of electrical sheet packages

- The individual sheets are fixed together with adhesive
- Integrated bonding processes such as Glulock® from Kienle & Spiess available on the market
- Partial or full-surface adhesive application possible
- Process must be adapted to the product (no adhesive leakage)



Graphic: Kienle + Spiess

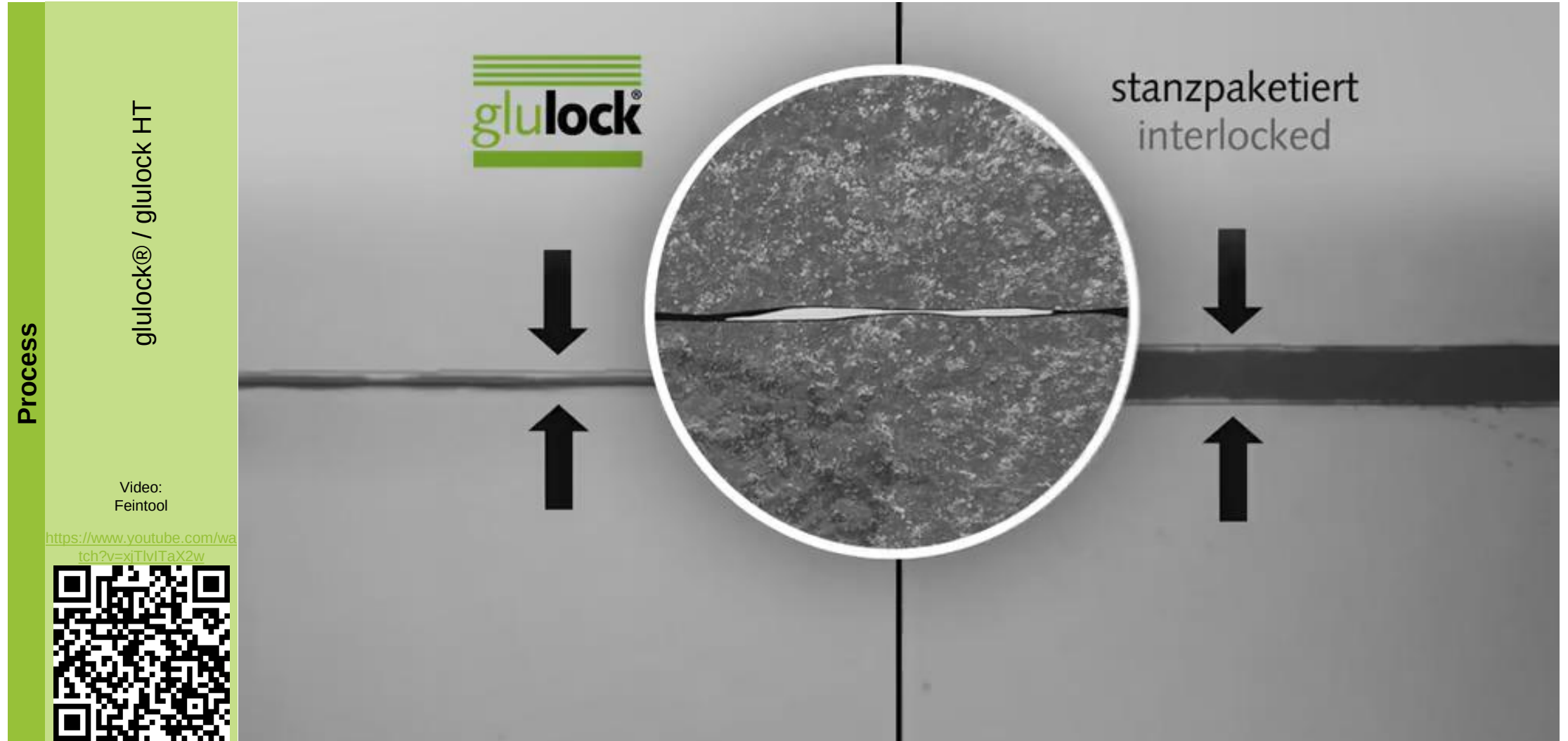
Advantages

- Can be used manually and automatically
- Processing of extremely thin sheets possible
- No influence on the magnetic flux
- No additional stacking and joining devices required with an integrated approach

Disadvantages

- Low mechanical strength for point-type adhesive joints
- Limited temperature resistance of the adhesive bond
- Aging of the adhesive (durability due to housing or other process)
- Long curing times until final strength is achieved

Individual sheets can also be glued together to form a sheet stack.



“Backlack” technology is suitable for highly efficient drives, as the magnetic flux is not disturbed.

Properties of bonding varnish for electrical steel laminations

- Application of the lacquer system already in the rolling mill
- Activation of the paint by temperature, pressure and time
- Integrated processes are already available on the market
- Process must be adapted to the product (no adhesive leakage)
- No space is required on the lamination stack for additional joining elements
- Particularly suitable for small series and prototype construction

Advantages

- Full-surface bond for high mechanical strength
- No interference with magnetic flux
- No additional joints to interfere with magnetic flux
- High thermal conductivity, liquid tightness and good NVH behavior



Graphic: CDWälzholz

Disadvantages

- Complex process technology and parameter setting
- Low cycle time due to the long curing phase
- High costs due to the manufacturing process
- Limited temperature resistance of the adhesive bond
- High energy requirements in production

Stamp stacking is suitable for the production of large quantities (> 100,000 sheet packages).

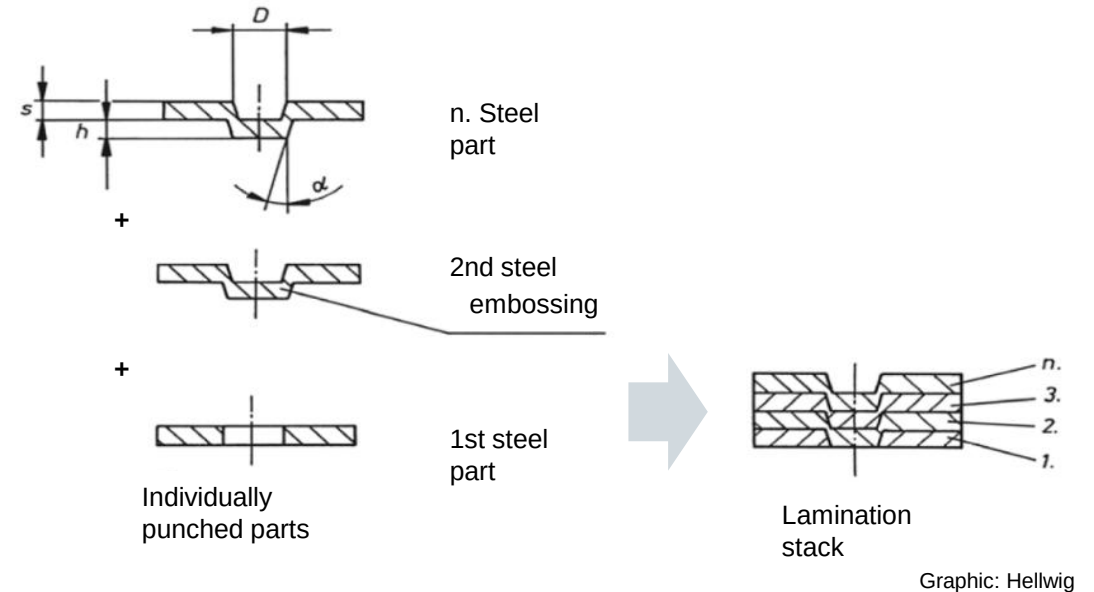
Interlocking of electrical steel laminations

- Production of the joint as the last step in the stamping process
- Introduction of interspersions into the material
- Fully integrated process in the subsequent cutting process
- Advanced tooling and process design



Advantages

- High productivity and output rate
- High degree of automation
- Unit costs are very low for large quantities
- No additional stacking and joining devices required



Disadvantages

- High tooling costs => not profitable for small quantities
- Interconnection points interfere with magnetic flux and can damage the insulation layer
- Very thin sheet thicknesses (<0.2 mm) cannot be processed
- No full surface bonding => low mechanical strength

The joining process (ElKuKlam) is a method for lamination stacking developed at the FAPS institute.

Electric sheet plastic clamp

Efficient for the production and assembly of lamination stacks for electrical machines

- Insertion of the joining elements by extrusion
- Improvement of the joint connection due to different sized joint holes
- Sophisticated tool and process design

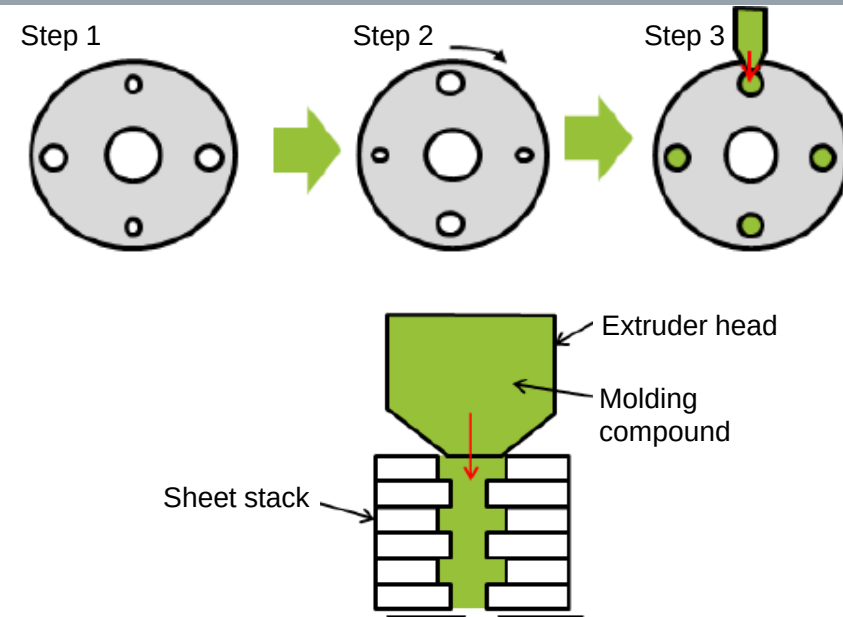
Process

- Step 1: Holes are varied asymmetrically
- Step 2: Blank is stacked and rotated 90
- Step 3: Plastic injection nozzle injects compound into cavity

Advantages

- Suitable for large-scale production
- Joints are not electrically conductive
- Combination with insulation process
- No additional stacking and joining devices required

Principle scheme:

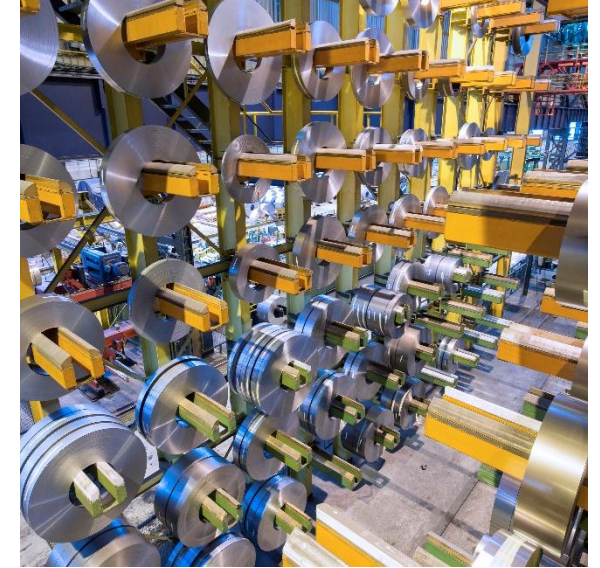


Disadvantages

- High tool costs (injection molding tool)
- Handling of loose sheets in injection molding tool difficult
- Additional recesses in the sheet metal required
- Temperature resistance of the joint

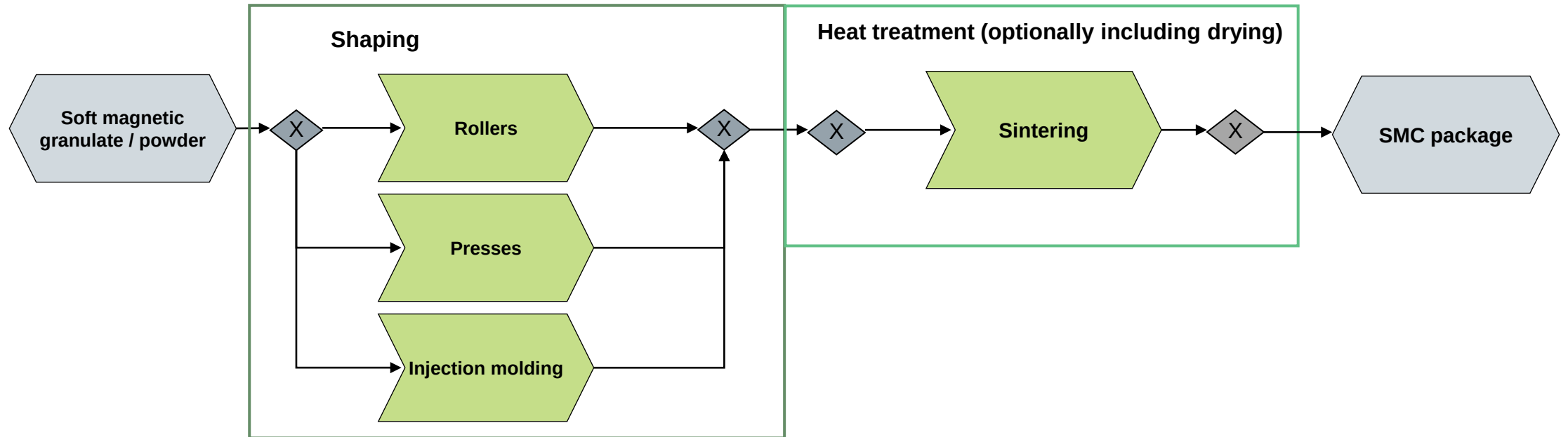
Lecture unit 2 provides an overview of the classification and manufacturing of soft magnetic materials.

- Overview and classification of soft magnetic materials (especially electrical steel)
- Production of electrical steel
- Insulating coatings for electrical sheets
- Processing of electrical steel (cutting, packaging)
- **Alternative soft magnetic core materials**



Graphic: ThyssenKrupp

The production of soft magnetic cores from powder composite materials is divided into the process steps of molding and heat treatment.



Compared to electrical steel, powder composite materials are characterized in particular by their three-dimensional design freedom.

Properties of soft magnetic composites (SMC)

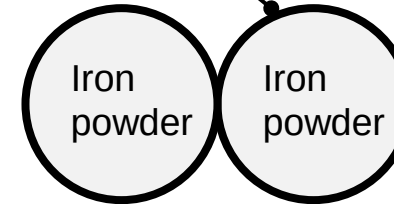
Efficient for the production and assembly of sheet metal packages for electrical machines

- Soft magnetic core is made from compressed iron powder
- High-purity iron powder has a special surface coating
- For use at high frequencies
- Ideal for three-dimensional flow control (e.g. transverse flow machine)



Graphic: GKN

Insulation layer



Graphic: PMG Füssen

Advantages

- Lower core losses at higher frequencies
- Lower electrical conductance
- Three-dimensional core structures with improved flow control
- Isotropic magnetic properties

Disadvantages

- Low permeability
- Low tensile strength
- High raw material price
- Complex and expensive process chain

Soft magnetic composites (SMC) enable high geometric flexibility within the production process.

Process

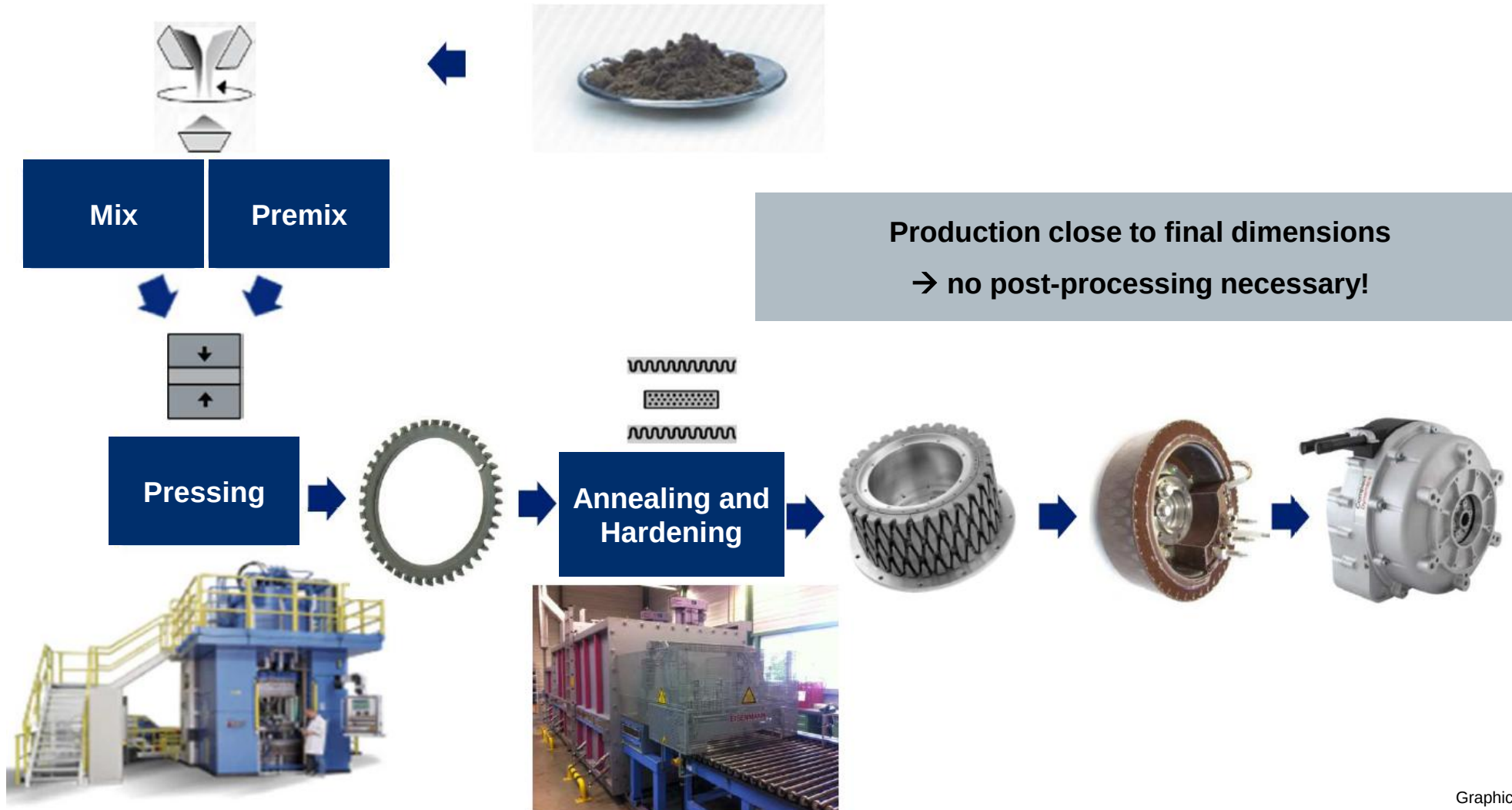
Soft Magnetic Composite /
Axial Flux Motors

Video:
Sumitomo Electric Group

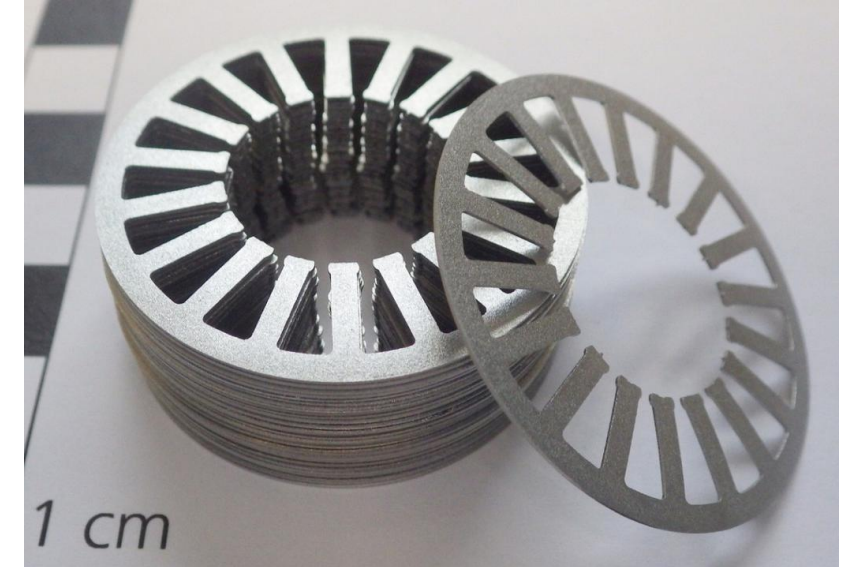
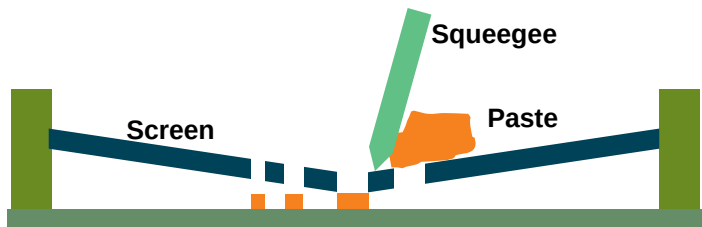
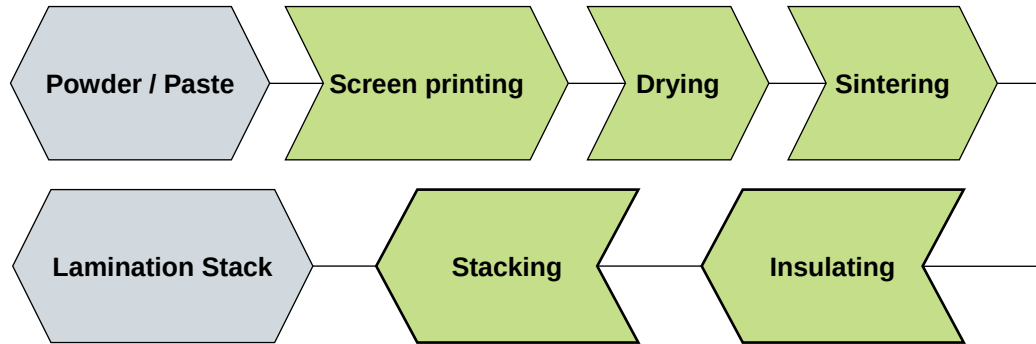
https://www.youtube.com/watch?v=IG_qZSX0J_c



The manufacturing processes of certain SMCs allow subsequent machining and can be used for prototypes.



Together with Siemens and other partners, Institute FAPS is involved in research on printed electrical sheets.



Graphic: IFAM Dresden

Advantages

- Thin electrical sheets
- Lower eddy current losses
- No waste
- Complex geometries possible

Disadvantages

- Variations in dimensional accuracy
- Running of the edges
- Handling of the green parts
- Insulation and bonding only possible after sintering

The contents of the lecture unit can be further deepened with the following specialized literature:

Books

- Boll, R., ed. *Soft magnetic materials. Introduction to magnetism, VAC materials and their applications*. 4th, completely revised and expanded ed. Berlin: Siemens-Aktienges. Dept. of Publishing, 1990. ISBN 3800915464
- Hellwig, W. *Chipless production: Punching. Conventional punching, high-performance punching, fine blanking*. 7th, completely revised edition. Wiesbaden: Vieweg+Teubner Verlag, 2001. from the programme Fundamentals of mechanical engineering. ISBN 9783528640422
- Callister, W., Rethwisch, D. *Materials Science and Engineering, An Introduction*. 1st edition. Weinheim: Wiley-VCH, 2012. ISBN 978-3-527-33007-2

Other publications

- Thyssen Krupp: Non-grain oriented electrical steel PowerCore®, product catalogue.
- Thyssen Krupp: Ways of production, Packaging Steel.
- Wuppermann, C.-D. and A. Schoppa. Leaflet 401. electrical steel strip and sheet, 2017. available at: https://www.stahl-online.de/wp-content/uploads/2019/04/MB401_Elektroband_und_blech.pdf.



The contents of the lecture unit can be further deepened with the following specialized literature:

Essays

- Bertotti, G. General properties of power losses in soft ferromagnetic materials [online]. *IEEE Transactions on Magnetics*, 1988, **24**(1), 621-630. ISSN 0018-9464. Available at: doi:10.1109/20.43994
- Tietz, M. and F. Herget. *Soft magnetic materials for electromobility (simulation and material)*. Stauseehalle, Mulfingen, 18 November 2011. FEMAG User Meeting 2011

Dissertations

- Lamprecht, E. *The influence of manufacturing processes on the eddy current losses of stator single-tooth laminations for use in hybrid and electric vehicles*. Dissertation. Bamberg: Meisenbach, 2014. production technology Erlangen. 247. ISBN 978-3-87525-362-7
- Veigel, M. *A new model for the calculation of production-dependent remagnetisation losses in synchronous machines*. Dissertation. Karlsruhe, 2018





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and Production Systems**



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THANK YOU